Category: a category $\mathcal{A}$ is:

Ab-category: if $\forall \mathrm{Hom}_{\mathcal{A}}(A,B)$ is given a str. of an abelian group,
(Preadditive) whose composition distributes over addition.

e.g. $A \xrightarrow{f} B \underset{g}{\overset{g'}{\rightrightarrows}} C \xrightarrow{h} D \Rightarrow h(g'+g)f = hg'f + hgf$

additive category: if $\mathcal{A}$ is an Ab-category w/ a zero object and product str.

① zero object: an object that is initial & terminal, denoted as 0 (unique up to equivalence)

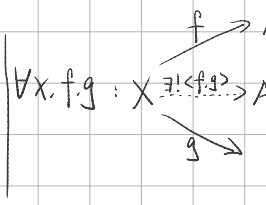
- initial object I: $\forall X \in Ob(\mathcal{A}), \exists! (i: I \to X) \in Mor(\mathcal{A})$
- terminal — T: —— , $\exists! (t: X \to T) \in Mor(\mathcal{A})$

$\Rightarrow$ Given $\forall X, Y \in Ob(\mathcal{A})$, we could construct their zero map through: $X \xrightarrow{t} 0 \xrightarrow{i} Y$

eg. for category Set: initial object: $\emptyset$ terminal object: $\{pt\}$, no zero object
$\Rightarrow$ not an additive category

② product str.:

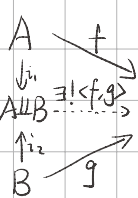
$\forall A, B \in Ob(\mathcal{A})$, define their product as: $A \times B$ w/ two projective morphism $p_1: A\times B \to A$, $p_2: A\times B \to B$,
(Rm. it seems like the Cartesian product)

$\forall X, f, g$: 

$\Rightarrow$ Generalize to a family of objects $\{A_\alpha\}_{\alpha \in I} \Rightarrow$ product: $\prod_{\alpha \in I} A_\alpha$

③ coproduct str.: the dual concept.

$\forall A, B \in Ob(\mathcal{A})$, define their coproduct as $A \amalg B$ w/ two inclusion morphism $i_1: A \to A\amalg B$, $i_2: B \to A \amalg B$,
(Rm. it seems like the non-insertion union)

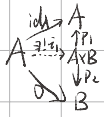
$\forall X, f, g$: 

$\Rightarrow$ Generalize to a family of objects $\{A_\alpha\}_{\alpha \in I} \Rightarrow$ coproduct: $\coprod_{\alpha \in I} A_\alpha$

In general category, coproduct $\neq$ product, however in additive category, the same (for finite product, not infinite, since Hom(A,B) abel group do not admit infinite sum as default.)

we shall prove that finite coproduct = finite product in additive category. through constructing the inclusion map w/ the help of zero map $A \to 0 \to B \Leftrightarrow 0: A \to B$

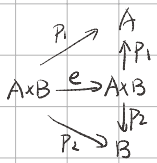
I. from the universal property of $A \times B$:



$\Rightarrow i_1 \checkmark$, similarly: $i_2 \checkmark$

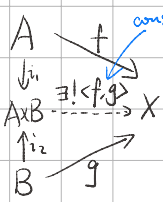
II. then we prove: $\underbrace{i_1 \circ p_1 + i_2 \circ p_2}_{=: e} = id_{A \times B}$

$\Rightarrow p_1 \circ e = p_1 \circ (i_1 \circ p_1 + i_2 \circ p_2) = p_1$, $p_2 \circ e = p_2 \Rightarrow$



through uniqueness $\Rightarrow e = id_{A \times B}$

III. prove universal property for coproduct.



construct as: $\langle f, g \rangle = f \circ p_1 + g \circ p_2 \Rightarrow$ diagram commutes

uniqueness: if $\exists h$ satisfies the same universal property.

$\Rightarrow h = h \circ id_{A \times B} = h \circ (i_1 \circ p_1 + i_2 \circ p_2) = f \circ p_1 + g \circ p_2 = \langle f, g \rangle$

□

Backwards: coproduct → product, similarly.

Thus, we may use the notation of biproduct: $A \oplus B$, $\bigoplus_{\alpha \in I} A_\alpha$

eg. 1) the right R-modules: mod-R is an additive category

2) the category of chain complexes: Ch is an additive category

nite coproducts. The zero object in **Ch** is the complex "0" of zero modules and maps. Given a family $\{A_\alpha\}$ of complexes of R-modules, the product $\prod A_\alpha$ and coproduct (direct sum) $\oplus A_\alpha$ exist in **Ch** and are defined degreewise: the differentials are the maps

$$\prod d_\alpha : \prod_\alpha A_{\alpha,n} \to \prod_\alpha A_{\alpha,n-1} \quad \text{and} \quad \oplus d_\alpha : \oplus_\alpha A_{\alpha,n} \to \oplus_\alpha A_{\alpha,n-1},$$

respectively. These suffice to make **Ch** into an additive category.

Exercise 1.2.1 Show that direct sum and direct product commute with homology, that is, that $\oplus H_n(A_\alpha) \cong H_n(\oplus A_\alpha)$ and $\prod H_n(A_\alpha) \cong H_n(\prod A_\alpha)$ for all n.

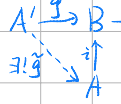
Before moving on, we redefine: kernel, cokernel, monic, epi in a category way.

In any additive category $\mathcal{A}$, consider a morphism $f: B \to C$

kernel: a map $i: A \to B$ s.t. $f \circ i = 0$ w/ the universal property:

$\forall A', \forall g: A' \to B$ s.t. $f \circ g = 0$

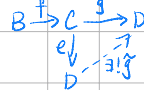
$\Rightarrow \exists! \tilde{g}: A' \to A$ s.t. commute



cokernel: dually: a map $e: C \to D$ s.t. $e \circ f = 0$ w/ universal property:

$\forall D', \forall g: C \to D'$ s.t. $g \circ f = 0$

$\Rightarrow \exists! \tilde{g}: D \to D'$ s.t. commute



monic: category version of injectivity $\Leftarrow$ left cancellation prop.

def: a map $i: A \to B$ is monic: $\forall g_1, g_2: A' \to A$ s.t. $i g_1 = i g_2$, then $\Rightarrow g_1 = g_2$

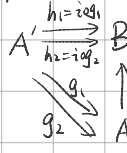
equivalently in additive category: $\forall g: A' \to A$, $ig = 0 \Rightarrow g = 0$

epi: dually: a map $e: C \to D$: $\forall g_1, g_2: D \to D'$ s.t. $g_1 e = g_2 e$, $\Rightarrow g_1 = g_2$

——— additive —: $\forall g: D \to D'$, $ge = 0 \Rightarrow g = 0$

every kernel is monic, dually, every cokernel is epi.

w/ the help of universal prop. of kernel, consider $h_1 = i \circ g_1$, $h_2 = i \circ g_2$, $h_1 = h_2$



$\Rightarrow f \circ h_1 = f \circ h_2 = 0$

$\Rightarrow \exists$ unique $\tilde{h}_1, \tilde{h}_2$: $h_1 = i \circ \tilde{h}_1$, $h_2 = i \circ \tilde{h}_2$

however: two candidates: $\tilde{h}_1 = g_1, g_2$

uniqueness $\Rightarrow g_1 = g_2$

Rm. the definition might counter our intuition from set.

e.g. the inclusion $i: \mathbb{Z} \hookrightarrow \mathbb{Q}$ is an epi in Ring, category of unital ring

$\forall R, \forall g, h: \mathbb{Q} \to R$ s.t. $g \circ i = h \circ i$ preserving the identity

$\Leftrightarrow \forall n \in \mathbb{Z}$, $g(n) = h(n)$,

we shall prove that: $\forall \frac{n}{m} \in \mathbb{Q}$, $g(\frac{n}{m}) = h(\frac{n}{m})$, obvious since g, h are ring homomorphisms.

The reason is: Ring is not even additive: let $f, g \in \mathrm{Hom}_{\mathrm{Ring}}(R, S)$

$(f+g)(ab) = f(a)f(b) + g(a)g(b)$

$(f+g)(a) \cdot (f+g)(b) = f(a)f(b) + g(a)g(b) + f(a)g(b) + f(b)g(a)$ $\Rightarrow \mathrm{Hom}_{\mathrm{Ring}}(R, S)$ is not abelian

Pre-abelian Category: requiring all morphism existing ker & coker, then:

Given $f: A \to B$, define two types of image: $\begin{cases} \mathrm{im}(f) = \ker(\mathrm{coker}(f)) \subseteq B \\ \mathrm{coim}(f) = \mathrm{coker}(\ker(f)) \subseteq A \end{cases}$

we would prove the canonical decomposition: $\exists! \tilde{f}$ s.t. $f = i \circ \tilde{f} \circ e$

$$\ker(f) \xrightarrow{i'} A \xrightarrow{f} B \xrightarrow{e'} \mathrm{coker}(f)$$
$$\downarrow e \qquad\qquad \uparrow i$$
$$\mathrm{coim}(f) \xrightarrow{\tilde{f}} \mathrm{im}(f)$$

① $A \xrightarrow{f} B \xrightarrow{e'} \mathrm{coker}(f) \Rightarrow f = i \circ f_1$

$\exists! f_1 \searrow \uparrow i$

$\mathrm{im}(f)$

② $\ker(f) \xrightarrow{i'} A \xrightarrow{f_1} \mathrm{im}(f)$ $\quad i$ is monic $\Rightarrow i \circ (f_1 \circ i') = (i \circ f_1) \circ i' = 0 \Rightarrow f_1 \circ i' = 0$

$\downarrow e \quad \nearrow \exists! f_2$ $\quad \Rightarrow f_1 = f_2 \circ e$ □

$\mathrm{coim}(f)$

To make it coincide w/ our sense to calculate homology et al.

Abelian Category: an additive category $\mathcal{A}$ s.t.

1. $\forall f \in \mathrm{Mor}(\mathcal{A})$, $\ker(f)$, $\mathrm{coker}(f)$ exists (Pre-abelian)
2. every monic f is the kernel of its cokernel: $f = \ker(\mathrm{coker}(f))$
3. every epi f is the cokernel of its kernel: $f = \mathrm{coker}(\ker(f))$

Long Exact Sequence.

Connecting homomorphism:

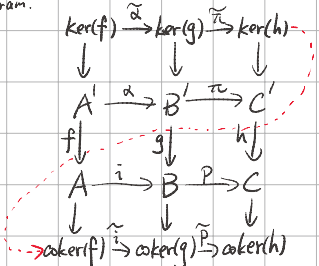
Let $0 \to A \to B \to C \to 0$ be a SES of chain complex $\Rightarrow$ natural maps $\partial: H_n(C) \to H_{n-1}(A)$ s.t. $\cdots \to H_{n+1}(C) \xrightarrow{\partial} H_n(A) \to H_n(B) \to H_n(C) \xrightarrow{\partial} H_{n-1}(A) \to \cdots$ is exact.

it's from the snake lemma, which is available in arbitrary abelian cat.

if the row is exact in the CD. (commutative diagram.)

$$\begin{array}{ccccccc} A' & \xrightarrow{\alpha} & B' & \xrightarrow{\pi} & C' & \to & 0 \\ \downarrow f & & \downarrow g & & \downarrow h & & \\ 0 & \to & A & \xrightarrow{i} & B & \xrightarrow{p} & C \end{array}$$

then:



$\tilde{\alpha} = \alpha|_{\ker(f)}$

$\tilde{\pi} = \pi|_{\ker g}$

$\tilde{i}([a]_f) = [i(a)]_g$

$\tilde{p}([b]_g) = [p(b)]_h$

$\Rightarrow$ exists exact sequence: $\ker(f) \xrightarrow{\tilde{\alpha}} \ker(g) \xrightarrow{\tilde{\pi}} \ker(h) \xrightarrow{\partial} \operatorname{coker}(f) \xrightarrow{\tilde{i}} \operatorname{coker}(g) \xrightarrow{\tilde{p}} \operatorname{coker}(h)$

where $\partial := i^{-1} \circ g \circ \pi^{-1}$

I. Verify the definition is fine.

① $\ker(f) \to \ker(g)$. take $x \in \ker f$. $\Rightarrow g(\alpha(x)) = i \circ f(x) = 0 \Rightarrow \alpha(x) \in \ker(g)$.

② $\operatorname{coker}(f) \to \operatorname{coker}(g)$, if we change $a \in A$ by $f(x) \in A \Rightarrow i(a + f(x)) = i(a) + g(\alpha(x)) \Rightarrow [i(a)]_g = [i(a + f(x))]_g$

II. Construct the connecting homomorphism.

Given $c' \in \ker(h)$. since $\pi: B' \to C'$ is surjective $\Rightarrow$ choose $b' \in B'$ s.t. $\pi(b') = c'$

from commutativity: $p(g(b')) = h(\pi(b')) = h(c') = 0 \Rightarrow g(b') \in \ker(p) \simeq \operatorname{im}(i)$ (exactness)

Since i is injective $\Rightarrow \exists! \, a \in A$ s.t. $i(a) = g(b') \Rightarrow a = i^{-1} \circ g \circ \pi^{-1}(c')$.

finally, the only redundance comes from the lift π^{-1}. if we change by $b_1' - b_2' \in \ker(\pi) = \operatorname{im}(\alpha)$, i.e.:

$b_1' - b_2' = \alpha(x')$, $x' \in A'$, then the corresponding a_1, a_2 satisfies:

$a_1 - a_2 = i^{-1}\left(g(b_1' - b_2')\right) = i^{-1}(g \circ \alpha(x')) = i^{-1}(i \circ f(x')) = f(x')$

therefore. to remove the redundancy of the lift π^{-1}, we shall focus on $A/\operatorname{im} f \simeq \operatorname{coker}(f)$

$\Rightarrow \partial: \ker h \to \operatorname{coker} f$ is constructed & well-defined.

Ⅲ verify the exactness.

① $\ker(f) \xrightarrow{\widetilde{\alpha}} \ker(g) \xrightarrow{\widetilde{\pi}} \ker(h)$: $\widetilde{\pi}\circ\widetilde{\alpha}=0$ is straightforward $\Rightarrow \operatorname{im}\widetilde{\alpha} \le \ker\widetilde{\pi}$

take $b' \in \ker\widetilde{\pi} \le \ker g$ s.t. $\widetilde{\pi}(b')=\pi(b')=0 \Rightarrow \exists\, a' \in A'$ s.t. $b'=\alpha(a')$

since $0=g(b')=g\circ\alpha(a')=\tilde{\imath}\circ f(a')$, $\tilde{\imath}$ injective $\Rightarrow a' \in \ker f$ □

② $\ker(g) \xrightarrow{\widetilde{\pi}} \ker(h) \xrightarrow{\partial} \operatorname{coker}(f)$

first verify $\partial\circ\widetilde{\pi}=0 \Rightarrow \operatorname{im}\widetilde{\pi} \le \ker\partial$

take $b' \in \ker(g)$, $c'=\pi(b')$. since $h\circ\pi(b')=p\circ g(b')=0 \Rightarrow c' \in \ker(h)$

therefore, we could directly use b' as the lift of c' when calculating $\partial c'$: $g(b')=\tilde{\imath}(a)=0 \Rightarrow a=0$

then verify $\ker\partial \le \operatorname{im}\widetilde{\pi}$

let $c' \in \ker(h)$ s.t. $\partial c'=0$, i.e. $\exists\, b' \in B'$, $a \in A$ s.t. $\pi(b')=c'$, $\tilde{\imath}(a)=g(b')$. $\underbrace{a=f(a')}_{[a]_f=0}$, $a' \in A'$

the question is that b' might not in $\ker(g)$, which is against the construction.

$\Rightarrow$ modify the lift as: $\widetilde{b'}=b'-\alpha(a')$, then: $g(\widetilde{b'})=\tilde{\imath}(a)-\underbrace{g\circ\alpha(a')}_{=\tilde{\imath}\circ f}=0$, while $\pi(\widetilde{b'})=\pi(b')=c'$ inv.

then: $c' \in \operatorname{im}(\widetilde{\pi}: \ker(g) \to \ker(h))$. □

③ $\ker(h) \xrightarrow{\partial} \operatorname{coker}(f) \xrightarrow{\widetilde{\imath}} \operatorname{coker}(g)$

let $\partial(c')=[a]_f$, i.e. $\exists\, b' \in B'$ s.t. $i(a)=g(b')$

$\Rightarrow \widetilde{\imath}\circ\partial(c')=\widetilde{\imath}([a]_f)=[i(a)]_g=[g(b')]_g=0 \Rightarrow \operatorname{im}\partial \le \ker\widetilde{\imath}$

backwards: take $[a]_f \in \operatorname{coker} f$ s.t. $\widetilde{\imath}([a]_f)=0$, i.e. $[i(a)]_g=0 \Rightarrow \exists\, b' \in B'$, $i(a)=g(b')$

then for $c'=\pi(b')$, $h(c')=h\circ\pi(b')=p\circ g(b')=0 \Rightarrow c' \in \ker h \Rightarrow \partial(c')=[a]_f$ □

④ $\operatorname{coker}(f) \xrightarrow{\widetilde{\imath}} \operatorname{coker}(g) \xrightarrow{\widetilde{p}} \operatorname{coker}(h)$

$\widetilde{p}\circ\widetilde{\imath}([a]_f)=[p\circ i(a)]_h=0 \Rightarrow \operatorname{im}\widetilde{\imath} \le \ker\widetilde{p}$

let $[b]_g \in \operatorname{coker} g$ s.t. $\widetilde{p}([b]_g)=0$, i.e. $p(b)=h(c')$, $c' \in C'$. we shall construct sth. related to $\operatorname{im}\tilde{\imath}=\ker p$

since $\pi: B' \to C'$ surjective $\Rightarrow \exists\, b' \in B'$ s.t. $\pi(b')=c'$

$\Rightarrow p(b-g(b'))=p(b)-p\circ g(b')=p(b)-h\circ\pi(b')=0 \Rightarrow b-g(b') \in \ker p \simeq \operatorname{im}\tilde{\imath}$

$\Rightarrow \exists\, a \in A$ s.t. $b-g(b')=\tilde{\imath}(a)$

$\Rightarrow [b]_g=[\tilde{\imath}(a)+g(b')]_g=\widetilde{\imath}[a]_f$ □

Rm. there are certain enhancement:

① if $\alpha: A' \to B'$ is monic, then so is the restriction $\widetilde{\alpha}: \ker(f) \to \ker(g)$

② if $p: B \twoheadrightarrow C$ is epi, then so is the quotient $\widetilde{p}: \operatorname{coker}(g) \to \operatorname{coker}(h)$

Applying to homology:

Prequisite: Given a complex C, consider such maps:

$\overline{d}_n: C_n/B_n(C) \to Z_{n-1}(C)$, $[c] \mapsto dc$ is well-defined. Then:

$\ker(\overline{d}_n) = \{c \in C_n \mid dc = 0\}/B_n(C) = Z_n(C)/B_n(C) = H_n(C)$.

$\operatorname{coker}(\overline{d}_n) = Z_{n-1}(C)/\operatorname{im}(\overline{d}_n) = Z_{n-1}(C)/B_{n-1}(C) = H_{n-1}(C)$

For every morphism $f: A \to B$, we have the following 2 SES.

$$\begin{cases} 0 \to \ker(f) \to A \to \operatorname{im}(f) \to 0 \\ 0 \to \operatorname{im}(f) \to B \to \operatorname{coker}(f) \to 0 \end{cases}$$

From the splicing: given 2 SES $\begin{cases} 0 \to K \to X \xrightarrow{\pi} M \to 0 \\ 0 \to M \xrightarrow{i} Y \to Q \to 0 \end{cases}$

we naturally obtain that: $0 \to K \to X \xrightarrow{i\circ\pi} Y \to Q \to 0$ is exact.

pf: since i injective $\Rightarrow \ker(i\circ\pi) = \ker(\pi) = \operatorname{im}(K \to X)$

left cancellation.

since π surjective $\Rightarrow \operatorname{im}(i\circ\pi) = \operatorname{im}(i) = \ker(Y \to Q)$

right cancellation

Thus, we obtain an exact sequence: $0 \to \ker(f) \to A \to B \to \operatorname{coker}(f) \to 0$, i.e. we have the exact sequence for the given complex C:

$$0 \to H(C) \to C/B(C) \to Z(C)[-1] \to H(C)[-1] \to 0.$$

Further, given $0 \to A_\cdot \xrightarrow{i} B_\cdot \xrightarrow{\pi} C_\cdot \to 0$, consider the diagram:

$$\begin{array}{ccccccc} 0 & \not\to & A_n/B_n(A) & \xrightarrow{\overline{i}} & B_n/B_n(B) & \xrightarrow{\widetilde{\pi}} & C_n/B_n(C) \to 0 \\ & & \downarrow \overline{d}_n^A & & \downarrow \overline{d}_n^B & & \downarrow \overline{d}_n^C \\ 0 & \to & Z_{n-1}(A) & \xrightarrow{\widetilde{i}} & Z_{n-1}(B) & \xrightarrow{\overline{\pi}} & Z_{n-1}(C) \not\to 0 \end{array}$$

verify: $\tilde{\pi}: B_n/B_n(B) \to C_n/B_n(B)$, $[b] \mapsto [\pi_n(b)]$

① well-define: from the commutativity: $\pi_{n-1}(d_B(b)) = d_C(\pi_n(b))$

② surjectivity: given $[c] \in C_n/B_n(B)$, $\exists b \in B_n$ s.t. $\pi_n(b) = c \Rightarrow \tilde{\pi}[b] = [\pi_n(b)] = [c]$

$\tilde{i}: Z_{n-1}(A) \to Z_{n-1}(B)$ well-definedness & injectivity is straight forward.

Rm: $\bar{i}: A_n/B_n(A) \to B_n/B_n(B)$ might not be injective.

if $[a] \in \ker \bar{i}$, i.e. $i(a) = d^B_{n+1} b$ for $b \in B_{n+1}$

however: $i(d^A_n a) = d^B_{n+1} \circ (i(a)) = d^B \circ d^B(b) = 0 \implies a \in Z_n(A) \Rightarrow [a] \in H_n(A)$

$\Rightarrow \ker \bar{i} \subseteq H_n(A)$ might be none zero.

further: let $c = \pi(b) \in C_{n+1} \Rightarrow d^C_{n+1}(c) = d^C_{n+1} \pi(b) = \pi \circ (d^B_{n+1} b) = \pi \circ i(a) = 0 \Rightarrow c \in Z_{n+1}(C)$

then: $\delta_{n+1}([c]) = [a] \in H_n(A) \Rightarrow \ker \bar{i} \subseteq \mathrm{Im}(\delta_{n+1}: H_{n+1}(C) \to H_n(A)) \subseteq H_n(A)$

and naturally: $\mathrm{Im}\, \delta_{n+1} \subseteq \ker \bar{i}$ through the construction of δ

$\Rightarrow \ker(\bar{i}: A_n/B_n(A) \to B_n/B_n(B)) \simeq \mathrm{Im}(\delta_{n+1}: H_{n+1}(C) \to H_n(A))$ might be none-zero.

similarly: $\bar{\pi}: Z_{n-1}(B) \to Z_{n-1}(C)$ might not be surjective.

take $c \in Z_{n-1}(C)$, lift to $b \in B_{n-1}$ s.t. $\pi(b) = c$. however:

$d_B b = i_{n-2}(a)$ w/ $\delta_{n-1}([c]) = [a]$ might be none-zero.

one may also prove that: $\mathrm{coker}(\bar{\pi}: Z_{n-1}(B) \to Z_{n-1}(C)) \simeq \mathrm{Im}(\delta_{n-1}: H_{n-1}(C) \to H_{n-1}(A))$

the above facts have the origin of adjoint functor POV. to be discussed later.

sketch: quotient $\Leftrightarrow$ cokernel $\Rightarrow$ colimit $\Rightarrow$ right exact

restriction $\Leftrightarrow$ kernel $\Rightarrow$ limit $\Rightarrow$ left exact.

Combine the 2 facts above, we finally proved the exactness of the LES ☐

Further, it could be generalized to arbitrary long SES. take an example of 5 terms:

$0 \to K \to A \xrightarrow{f} B \xrightarrow{g} C \to Q \to 0$

let $I_1 = \operatorname{im} f$, $I_2 = \operatorname{im} g$. then the above sequence breaks into:

$0 \to \ker f \to A \to \operatorname{im} f \to 0$

$0 \to \operatorname{im} f \to B \to \operatorname{coker} f \simeq \operatorname{im} g \to 0$

$0 \to \operatorname{im} g \to C \to \operatorname{coker} g \to 0$

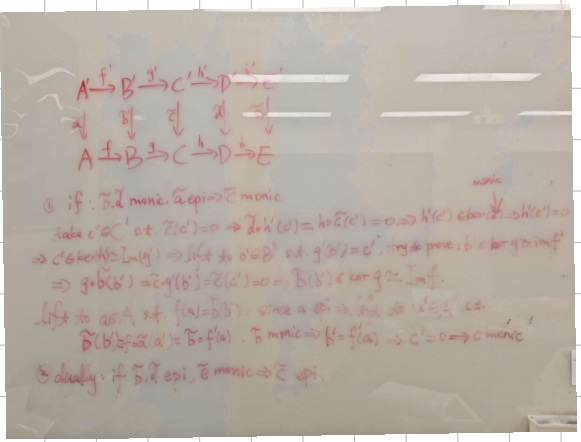
$K = \ker f$

$0 \to \underset{\operatorname{im} f}{\ker g} \to B \to \operatorname{im} g \to 0 \Rightarrow \operatorname{im} g \simeq \operatorname{coker} f$.

then apply to the 3 terms SES, then combine together. specifically:

$H_n(Q) \xrightarrow{\delta_1} H_{n-1}(I_2) \xrightarrow{\delta_2} H_{n-2}(I_1) \xrightarrow{\delta_3} H_{n-3}(K)$ is the connecting homomorphism.

Ex. 5-lemma



Ex 1. $0 \to Z \to C \to B[-1] \to 0 \Rightarrow \cdots \to H_n(Z(C)) \to H_n(C) \to H_n(B(C)[-1]) \xrightarrow{\delta_n} H_n(Z(C))$

since $d(Z(C)) = 0$. $d(B(C)) = 0 \Rightarrow H_n(Z(C)) = Z_n(C)$, $H_n(B(C)[-1]) = B_{n-1}(C)$.

$\Rightarrow \cdots \to Z_n(C) \to H_n(C) \to B_{n-1}(C) \xrightarrow{\delta_n} Z_{n-1}(C) \to H_n(C) \to \cdots$

since $b \in B_{n-1}(C)$ is bdry $\Rightarrow b = dc$, the action of δ_n is lift then differential.

$\Rightarrow \operatorname{im} \delta_n = 0 \Rightarrow$ the LES breaks up into SES: $0 \to Z_n(C) \to H_n(C) \to B_{n-1}(C) \to 0$

Ex 2. let $f: A_\cdot \to B_\cdot$. if $K := \ker f$, $Q := \operatorname{coker} f$ are acyclic. i.e. $H_n(K) = H_n(Q) = 0, \forall n$

consider that: $0 \to K \to A \to B \to Q \to 0$

Chain Homotopy

Recall our decomposition of the complex chain of vector spaces as: $H_n' \cong H_n(C)$

$C_n = Z_n \oplus L_n$, $Z_n = B_n \oplus H_n'$, where $d_n|_{L_n}: L_n \xrightarrow{\sim} B_{n-1} \subseteq C_{n-1}$, i.e. $B_n = \mathrm{im} d_{n+1}$, $Z_n = \ker d_n$,

then we may achieve the identification: $C_n \cong B_n \oplus H_n(C) \oplus B_{n-1}$, uncanonically.

further, we may form another map $s: C_n \to Z_n \to B_n \xrightarrow{(d_{n+1}|_{L_{n+1}})^{-1}} L_{n+1} \hookrightarrow C_{n+1}$ then: $\underline{d = dsd}$

once we write $x \in C_n$ as: $x = b + h + l$, $b \in B_n$, $h \in H_n'$, $l \in L_n$

$(ds)(x) = d(s(b)) = b$, $(sd)(x) = s(d(l)) = l$

further: $(ds)^2 = dsds = ds$, $(sd)^2 = sdsd = sd$, $[ds, sd] = dssd - sdsd = ds^2 d$

$\Rightarrow$ the projective operator: $ds: C_n \to B_n$, $sd: C_n \to L_n$ under the condition: $s^2 = 0$

then: $ds + sd = id - \pi_H$, $\pi_H: C_n \to H_n$

thus: $ds + sd = id \iff C$ is exact & $d = dsd$

Now we fix a patch: without vector space decomposition and assumption of $ds^2 d = 0$

Lem. Given s s.t. $d = dsd$ and C exact, we may construct t s.t. $dt + td = id$

Motivation: $d = dsd$ means that: $\forall b \in B$, $ds|_B = id$. but it says nothing about L_n.

to extend s from $d^{-1}|_{B_n}: B_n \to L_{n+1}$, we shall kill all possible none-zero action on L_n.

$\Rightarrow$ construct $t := s \cdot (\underbrace{id - sd}_{\text{projection to } B_n, \text{ zero on } L_n})$

further: $t|_{B_n} = s|_{B_n}$, $d = dtd$, $t^2 = 0 \Rightarrow [dt, td] = 0 \Rightarrow$ commuting idempotents.

Rm: One may concern that s itself will send $B_n \to L_{n+1} \oplus B_{n+1}$, but $\mathrm{im}(s) \cong L_{n+1} \Rightarrow$ change of basis.

thus: since $\ker(td) \cong L_n$, $\mathrm{im}(td) \cong B_n \Rightarrow td + dt = id$. □

Definition: ① A complex C is called split if there are splitting maps $s_n: C_n \to C_{n+1}$ s.t. $d = dsd$

② In addition if C is exact(acyclic) $\Rightarrow C$ is split exact.

Rm: There exists complex that is exact but not split exact, i.e. no map s s.t. $ds + sd = id$ and decomposition: $C_n \cong Z_n \oplus B_n$

e.g: $\cdots \xrightarrow{\times 2} \mathbb{Z}/4 \xrightarrow{\times 2} \mathbb{Z}/4 \xrightarrow{\times 2} \mathbb{Z}/4 \to \cdots$

some exercise to be done:

Exercise 1.4.1 The previous example shows that even an acyclic chain complex of free R-modules need not be split exact.

1. Show that acyclic *bounded below* chain complexes of free R-modules are always split exact.
2. Show that an acyclic chain complex of finitely generated free abelian groups is always split exact, even when it is not bounded below.

Exercise 1.4.2 Let C be a chain complex, with boundaries B_n and cycles Z_n in C_n. Show that C is split if and only if there are R-module decompositions $C_n \cong Z_n \oplus B'_n$ and $Z_n = B_n \oplus H'_n$. Show that C is split exact iff $H'_n = 0$.

Chain Homotopy. Given 2 chain complexes $C_\cdot, D_\cdot$ w/ chain map $f: C \to D$

if there are maps (morphism) $s_n: C_n \to D_{n+1}$ s.t. $f = ds + sd$

$\Rightarrow f$ is called null homotopic, $\{s_n\}$ are called a chain contraction of f.

$$\to C_{n+1} \xrightarrow{d} C_n \xrightarrow{d} C_{n-1} \to$$
$$\downarrow s \quad \downarrow f \quad \downarrow s$$
$$\to D_{n+1} \xrightarrow{d} D_n \xrightarrow{d} D_{n-1} \to$$

then: $f_*: H_n(C) \to H_n(D)$ is a zero map.

pf: choose $[z] \in H_n(C) \Rightarrow f_*([z]) = [d(s([z]))] = 0$

two chain maps $f, g: C \to D$ are chain homotopic if $f - g$ is null homotopic. i.e.

if: $\exists \{s_n\}$ the chain homotopy from f to g s.t. $f - g = sd + ds$

then: $f_* = g_*$

$$Ch \xrightarrow{Q} K \xrightarrow{\bar{H}_n} R\text{-Mod}$$

s.t. $H_n = \bar{H}_n \circ Q$ i.e. well-defined.

Exercise 1.4.5 In this exercise we shall show that the chain homotopy classes of maps form a quotient category $\mathbf{K}$ of the category $\mathbf{Ch}$ of all chain complexes. The homology functors H_n on $\mathbf{Ch}$ will factor through the quotient functor $\mathbf{Ch} \to \mathbf{K}$.

1. Show that chain homotopy equivalence is an equivalence relation on the set of all chain maps from C to D. Let $\mathrm{Hom}_{\mathbf{K}}(C, D)$ denote the equivalence classes of such maps. Show that $\mathrm{Hom}_{\mathbf{K}}(C, D)$ is an abelian group.

   { reflexivity, symmetry, transitivity.

2. Let f and g be chain homotopic maps from C to D. If $u: B \to C$ and $v: D \to E$ are chain maps, show that vfu and vgu are chain homotopic. Deduce that there is a category $\mathbf{K}$ whose objects are chain complexes and whose morphisms are given in (1).

   left & right ideals

3. Let f_0, f_1, g_0, and g_1 be chain maps from C to D such that f_i is chain homotopic to g_i $(i = 1, 2)$. Show that $f_0 + f_1$ is chain homotopic to $g_0 + g_1$. Deduce that $\mathbf{K}$ is an additive category, and that $\mathbf{Ch} \to \mathbf{K}$ is an additive functor.

   Ab-cat.: Hom is abelian
   additive cat: zero object & finite biproduct
   abelian cat: ker/coker.

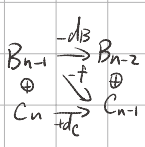
4. Is $\mathbf{K}$ an abelian category? Explain.

   Not.

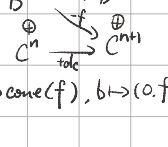
Mapping Cones & Cylinders: to encode information of $f: B \to C$ into differential

Given $f: B_\bullet \to C_\bullet$, construct a complex $\mathrm{cone}(f)$ s.t.

$\mathrm{cone}(f)_n = B_{n-1} \oplus C_n$, $d: (b,c) \mapsto (-d_B(b), d_C(c) - f(b))$: $\begin{bmatrix} -d_B & 0 \\ -f & +d_C \end{bmatrix}$



dually: for cochain complex: $\mathrm{cone}(f)^n = B^{n+1} \oplus C^n$ $d: \begin{bmatrix} -d_B & 0 \\ -f & +d_C \end{bmatrix}$



Prop: denote the embedding $i: C \hookrightarrow \mathrm{cone}(f)$, $c \mapsto (0,c)$, we have: if $f: B \to \mathrm{cone}(f)$, $b \mapsto (0, f(b))$

by constructing $s_n: B_n \to \mathrm{cone}(f)_{n+1}$, $b \mapsto (-b, 0)$

$\Rightarrow d_{\mathrm{cone}} \circ s(b) = d_{\mathrm{cone}}(-b, 0) = (d_B b, f(b))$, $s \circ d_B = (-d_B b, 0)$

$\Rightarrow i \circ f = ds + sd \Rightarrow i \circ f \simeq 0$

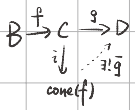
The construction is "similar" to the cone mapping in algebraic topology: $X \xrightarrow{f} Y \to C_f = CX \cup_f Y$.

Further, it gives a construction regarding the cokernel in the quotient cat. K.

universal property:

Given $f: B \to C$, $i: C \hookrightarrow \mathrm{cone}(f)$, $\forall$ complex D, $g: C \to D$ s.t. $g \circ f \simeq 0$

$\Rightarrow \exists \bar{g}: \mathrm{cone}(f) \to D$ s.t. $g \simeq \bar{g} \circ i$ in K



pf: $gf \simeq 0 \Rightarrow \exists$ chain contraction $\{s_n: B_n \to D_{n+1}\}$ s.t. $gf = sd_B + d_D s$

$\to$ construct $\bar{g}: \mathrm{cone}(f) \to D$, $(b,c) \mapsto s(b) + g(c)$

uniqueness does not follow since $\bar{g}$ could be arbitrary on $\mathrm{cone}(f)|_B$. $\Rightarrow$ weak cokernel

weak kernel is the homotopy fiber: $fib(f) = \Sigma^{-1}\mathrm{cone}(f) = \mathrm{cone}(f)[1]$, to be discussed in triangulated cat

Exercise 1.5.1 Let cone(C) denote the mapping cone of the identity map id_C of C; it has $C_{n-1} \oplus C_n$ in degree n. Show that cone(C) is split exact, with $s(b, c) = (-c, 0)$ defining the splitting map.

just to check that $ds + sd = id$.

Besides, we may construct the following SES:

$$0 \to C \to \mathrm{cone}(f) \xrightarrow{\delta} B[-1] \to 0, \quad \delta(b,c) = -b.$$

which yields: $\to H_{n+1}(\mathrm{cone}(f)) \xrightarrow{\delta_*} H_n(B) \xrightarrow{\partial} H_n(C) \to H_n(\mathrm{cone}(f)) \xrightarrow{\delta_*} H_{n-1}(B) \xrightarrow{\partial} \cdots$

$\qquad\qquad\qquad

Mapping Cylinder: $\mathrm{cyl}(f)_n = B_n \oplus B_{n-1} \oplus C_n$ w/:

$$d = \begin{pmatrix} d_B & \mathrm{id}_B & \\ & -d_B & \\ & -f & d_C \end{pmatrix} :$$

$$\begin{array}{ccccc} B_n & \xrightarrow{+d_B} & B_{n-1} & & \\ \oplus & \overset{+\mathrm{id}_B}{\nearrow} & \oplus & & \\ B_{n-1} & \xrightarrow{-d_B} & B_{n-2} & & \\ \oplus & \searrow^{-f} & \oplus & & \\ C_n & \xrightarrow{d_C} & C_{n-1} & & \end{array}$$

Lemma 1.5.6 *The subcomplex of elements $(0, 0, c)$ is isomorphic to C, and the corresponding inclusion $\alpha: C \to \mathrm{cyl}(f)$ is a quasi-isomorphism.*

$d[(b,b',c)] = [(d_B b + b', -d_B b', d_C c - f(b'))] \simeq (-d_B b', d_B b + b')$. $-(-\mathrm{id})$.

Proof The quotient $\mathrm{cyl}(f)/\alpha(C)$ is the mapping cone of $-\mathrm{id}_B$, so it is null-homotopic (exercise 1.5.1). The lemma now follows from the long exact homology sequence for

$$0 \longrightarrow C \xrightarrow{\alpha} \mathrm{cyl}(f) \longrightarrow \mathrm{cone}(-\mathrm{id}_B) \longrightarrow 0. \qquad \diamond$$

Exercise 1.5.4 Show that $\beta(b, b', c) = f(b) + c$ defines a chain map from $\mathrm{cyl}(f)$ to C such that $\beta\alpha = \mathrm{id}_C$. Then show that the formula $s(b, b', c) = (0, b, 0)$ defines a chain homotopy from the identity of $\mathrm{cyl}(f)$ to $\alpha\beta$. Conclude that α is in fact a chain homotopy equivalence between C and $\mathrm{cyl}(f)$.

in a word, given $f: B \to C \Rightarrow \mathrm{cyl}(f) \simeq C$ in $K = \mathrm{ch}/Q$, as expected in top. POV

Now we conclude for the mapping cone & the mapping cylinder.

In $\mathrm{cyl}(f)$, quotient out the subcomplex $\{(b,0,0)\} \simeq B$ gives: $\mathrm{cyl}(f)/B \simeq \mathrm{cone}(f)$

and: $B \to \mathrm{cyl}(f) \xrightarrow{\beta} C$ is just f, therefore:

$f_*: H(B) \to H(C)$ factors through $H(B) \to H(\mathrm{cyl}(f)) \to H(C)$.

written in commutative diagram:

$$\begin{array}{ccccccccc} & & & & C & & & & \\ & & & \overset{f}{\nearrow} & \uparrow{\scriptstyle \beta} & & & & \\ 0 & \longrightarrow & B & \longrightarrow & \mathrm{cyl}(f) & \longrightarrow & \mathrm{cone}(f) & \longrightarrow & 0 \\ & & & & \uparrow{\scriptstyle \alpha} & & \| & & \\ 0 & \longrightarrow & C & \longrightarrow & \mathrm{cone}(f) & \xrightarrow{\delta} & B[-1] & \longrightarrow & 0. \end{array}$$

The homology long exact sequences fit into the following diagram:

$$\begin{array}{ccccccccc} \cdots & \xrightarrow{\partial} & H_n(B) & \to & H_n(\mathrm{cyl}(f)) & \to & H_n(\mathrm{cone}(f)) & \xrightarrow{\partial} & H_{n-1}(B) & \to & \cdots \\ & & \| & \nearrow & \| & & \| & & \| & & \\ \cdots & \to & H_{n+1}(B[-1]) & \xrightarrow{\partial} & H_n(C) & \to & H_n(\mathrm{cone}(f)) & \xrightarrow{\delta} & H_n(B[-1]) & \xrightarrow{\partial} & \cdots \end{array}$$

Finally we state that: the weak cokernel $\mathrm{cone}(f)$ agree w/ the ordinary cokernel at the homology level. i.e.

consider: $f: B \to C$, if f is also part of a SES. says: $0 \to B \xrightarrow{f} C \xrightarrow{g} D \to 0$ ($D \cong \operatorname{coker} f$)

then: $H_n(D) \simeq H_n(\operatorname{cone}(f))$ is quasi-isomorphism, but not generally homotopy equivalent.

consider:
$$\begin{array}{ccccccccc} 0 & \to & B & \to & \operatorname{cyl}(f) & \to & \operatorname{cone}(f) & \to & 0 \\ & & \downarrow id_B & \circlearrowright & \downarrow \beta & \circlearrowright & \downarrow \varphi & & \\ 0 & \to & B & \xrightarrow{f} & C & \xrightarrow{g} & D & \to & 0 \end{array} \qquad \varphi(b,c) := g(c)$$

in previous text: β is a homotopy equivalence.

then consider the LES:

$$\begin{array}{ccccccccccc} H_n(B) & \to & H_n(\operatorname{cyl}(f)) & \to & H_n(\operatorname{cone}(f)) & \xrightarrow{\delta} & H_{n-1}(B) & \to & H_{n-1}(\operatorname{cyl}(f)) \\ \downarrow id_{B*} & & \downarrow \beta_* & & \downarrow \varphi_* & & \downarrow id_{B*} & & \downarrow \beta_* \\ H_n(B) & \to & H_n(C) & \to & H_n(D) & \xrightarrow{\delta} & H_{n-1}(B) & \to & H_{n-1}(C) \end{array}$$

From the 5-lemma $\Rightarrow \varphi_*$ is an isomorphism.

However, φ is not always chain equivalence.

Lemma: additive functor preserves exactness for split exact sequence

pf: Given $0 \to A \xrightarrow{i} B \underset{p}{\overset{s}{\rightleftarrows}} C \to 0$ split $\Rightarrow \exists s$ s.t. $p \circ s = id_C$. $B = A \oplus C \Rightarrow \exists r: B \to A$ s.t. $r \circ i = id_A$

$\Rightarrow i \circ r + s \circ p = id_B$

additive $\Rightarrow F(i) \circ F(r) + F(s) \circ F(p) = id_{F(B)}$

$\Rightarrow 0 \to F(A) \to F(B) \to F(C) \to 0$ is also split exact.

More about the Abelian Categories.

Def.

(1) a category $\mathcal{A}$ is **small** when $Ob(\mathcal{A})$ & $Mor(\mathcal{A})$ are set instead of class.

Rm. the difference is about the Russell paradox

in a word: a set could be an element of another set, but class is the union discribed by properties: $C=\{x \mid \varphi(x)\}$.

e.g. $C=\{\text{all sets}\}$ is a proper class, otherwise $C\in C$ is forbiddened. Cat. of Set is not small

local small: for arbitrary fixed object X,Y, $\mathrm{Hom}_{\mathcal{A}}(X,Y)$ is a set instead of class $\Rightarrow$ Set is local small

(2) a functor $E:\mathcal{A}\to R\text{-Mod}$ is **faithful** when distinct morphism remain distinct.

— **full** when $\forall A,B\in Ob(\mathcal{A})$, $\mathrm{Hom}_R(E(A),E(B))$ comes from $\mathrm{Hom}_{\mathcal{A}}(A,B)$

— **exact** when it preserves exact sequence.

Rm. when talking about a subcategory is full etc., we are consider the restriction functor

Theorem to be proved: Freyd-Mitchell embedding thm:

If $\mathcal{A}$ is a small abelian category,

then: there are a ring R & an exact fully faithful functor $E:\mathcal{A}\to R\text{-Mod}$. (Mod: large but local small.)

Lemma: Closure criterion for a full subcategory C of an abelian category $\mathcal{A}$

(1) C is additive $\Leftrightarrow$ C contains a zero object & closed under finite direct sum (biproduct)

(2) C is abelian and the inclusion is exact $\Leftrightarrow$ C is additive & C is closed under ker & coker (ambient)

↳ preserving ker & coker.

use examples in the following lecture to test. (consider the inclusion $j: Sh(X)\to PSh(X)$)

Functor categories:

Let $\mathcal{C}$ be a small category & $\mathcal{A}$ an abelian category.

Define the functor category: $\mathcal{A}^{\mathcal{C}}$ w/ objects: functors $F:\mathcal{C}\to\mathcal{A}$

and morphisms: natural transformation $\eta: F\Rightarrow G$

natural transformation: a family of maps $\eta_c: F(c)\to G(c)$, $\forall c\in Ob(\mathcal{C})$ s.t.

$$\forall\, u: c\to c' \Rightarrow \begin{array}{ccc} F(c) & \xrightarrow{F(u)} & F(c') \\ \downarrow{\scriptstyle \eta_c} & & \downarrow{\scriptstyle \eta_{c'}} \\ G(c) & \xrightarrow{G(u)} & G(c') \end{array} \qquad \begin{array}{c} F \\ \Downarrow \eta \\ G \end{array}$$

$Mor(\mathcal{C})=\emptyset$

e.g. $\mathcal{C}$ the discrete category of integers $\Rightarrow Ab^{\mathcal{C}}$ contains category of graded abelian group as subcat.

poset category of integers $(\cdots\to n\to n+1\to\cdots)\Rightarrow \mathcal{A}^{\mathcal{C}}$ contains $Ch(\mathcal{A})$ as abelian cat. of cochain complex

Prop. if $\mathcal{A}$ is abelian & $\mathcal{C}$ is small $\Rightarrow \mathcal{A}^{\mathcal{C}}$ is abelian

Pf: additive is inherited. For a natural transformation $\eta: F \Rightarrow G$. construct:

$\ker(\eta) =: K$ s.t. $K(C) = \ker(\eta_C)$ w/ kernel inclusion $k_C: K(C) \hookrightarrow F(C) \Rightarrow \eta_C k_C = 0$

Naturality $\Rightarrow \eta_{C'} \underbrace{F(u) k_C}_{f} = G(u) \eta_C k_C = 0$,

universal property: $\eta_{C'} f = 0 \Rightarrow \exists! k(u)$ s.t. $f = F(u) k_C = k_{C'} k(u)$.

$K(C) \xrightarrow{f} F(C') \xrightarrow{\eta_{C'}} G(C')$, $K(C') \xrightarrow{k_{C'}}$, $k(u): K(C) \to K(C')$

$\Rightarrow K$ is indeed a functor: $\mathcal{C} \to \mathcal{A} \Rightarrow K \in Ob(\mathcal{A}^{\mathcal{C}})$

canonical coIm-im isomorphism ($f = \ker(\text{coker} f)$ etc.) is obtained pointwise. $\Rightarrow \mathcal{A}^{\mathcal{C}}$ is abelian.

Rm. the key point is that these structure maps (morphisms) are not extra data but controlled

e.g. let $\mathcal{C}$ be the poset of open subset of X. $\Rightarrow$ the category of presheaves on X w/ values in $\mathcal{A}$:

$PSh(X) := \mathcal{A}^{\mathcal{C}^{op}}$ is an abelian category.

Def: let $F: \mathcal{A} \to \mathcal{B}$ an additive functor. given a SES $0 \to A \to B \to C \to 0$, F is called:

left exact if $0 \to F(A) \to F(B) \to F(C)$ / right exact if $F(A) \to F(B) \to F(C) \to 0$

A contravariant functor F is called left exact if $\tilde{F}: \mathcal{A}^{op} \to \mathcal{B}$ is left exact

Prop. $\mathrm{Hom}(M, -)$ is left exact.

pf: Given $0 \to A \xrightarrow{f} B \xrightarrow{g} C$ exact. we prove: $0 \to \mathrm{Hom}(M,A) \xrightarrow{f_*} \mathrm{Hom}(M,B) \xrightarrow{g_*} \mathrm{Hom}(M,C)$ is exact.

① choose $\alpha \in \mathrm{Hom}(M,A)$. then $f_* \alpha = f \circ \alpha$.

Since $f = \ker(g)$ is monic. once $f_* \alpha = f \circ \alpha = 0$ we have $\alpha = 0 \Rightarrow f_*$ is monic then is injective

② Similarly. one have $g_* f_* = 0$. one shall verify the universal property for kernel:

$\forall \beta \in \mathrm{Hom}(M,B)$ s.t. $g_* \beta = g \circ \beta = 0$. From the universal property for $0 \to A \xrightarrow{f} B \xrightarrow{g} C$

$\Rightarrow \exists! \alpha: M \to \alpha$ s.t. $\beta = f \circ \alpha$, i.e. $\beta = f_* \alpha \Rightarrow \ker g_* = \mathrm{im} f_*$

Rm. the factor property of kernel guaranteen the left exactness

the surjection property is not the dual proposition: Given $u: C \to M$ s.t. $u \circ g = 0 \Rightarrow u = 0 \Rightarrow g^*$ is monic.

the correct dual one is $\mathrm{Hom}_{\mathcal{A}}(-, M) = \mathrm{Hom}_{\mathcal{A}^{op}}(M, -)$ is contravariantly left exact

Yoneda Lemma: define $h: \mathcal{A} \to \mathcal{A}b^{\mathcal{A}^{op}}$, $A \mapsto \mathrm{Hom}_{\mathcal{A}}(-, A)$. then h reflects exactness. i.e.

$A \xrightarrow{\alpha} B \xrightarrow{\beta} C$ in $\mathcal{A}$ is exact, provided that $\forall M \in \mathcal{A}$. $\mathrm{Hom}_{\mathcal{A}}(M,A) \xrightarrow{\alpha_*} \mathrm{Hom}_{\mathcal{A}}(M,B) \xrightarrow{\beta_*} \mathrm{Hom}_{\mathcal{A}}(M,C)$ is exact.

$h: \mathcal{A} \to Ab^{\mathcal{A}^{op}}$. $A \mapsto h_A := \mathrm{Hom}_{\mathcal{A}}(-, A)$.

Given $u: M \to N$. $h_A(u): h_A(N) = \mathrm{Hom}_{\mathcal{A}}(N, A) \longrightarrow \mathrm{Hom}_{\mathcal{A}}(M, A)$, $\alpha \mapsto \alpha \circ u$.

Verify the natural transformations between h_A & h_B: define η according to $\mathrm{Hom}_{\mathcal{L}}(A, B)$.

$\forall u: C \to C'$ in $\mathcal{A}$ $\Rightarrow$

$$\begin{array}{ccc} \mathrm{Hom}_{\mathcal{A}}(C', A) & \xrightarrow{h_A(u)} & \mathrm{Hom}_{\mathcal{A}}(C, A) \\ \downarrow \eta_{C'} & & \downarrow \eta_C \\ \mathrm{Hom}_{\mathcal{A}}(C', B) & \xrightarrow{h_B(u)} & \mathrm{Hom}_{\mathcal{A}}(C, B) \end{array}$$

pf: let $M = A \Rightarrow \beta\alpha = \beta_* \alpha_* (id_A) = 0$

let $M = \ker(\beta)$ w/ $\iota: \ker(\beta) \to B$, then: $\beta_*(\iota) = \beta \circ \iota = 0$

since $\ker \beta_* = \mathrm{im}\, \alpha_* \Rightarrow \exists\, \delta \in \mathrm{Hom}(M, A)$ s.t. $\iota = \alpha_*(\delta) = \alpha \circ \delta \Rightarrow \ker(\beta) = \mathrm{im}(\iota) \subseteq \mathrm{im}(\alpha) \Rightarrow \ker\beta = \mathrm{im}\,\alpha$

⇒ Yoneda Embedding

Every additive category $\mathcal{A}$ can be embedded in the abelian category $Ab^{\mathcal{A}^{op}}$ (when $\mathcal{A}$ is abelian hence we could discuss exactness:) Since $\mathrm{Hom}_{\mathcal{A}}(M, -)$ is left exact

$\to \mathcal{A}$ is embedded in the subcategory $\mathcal{L}$ of all contravariant left exact functors: $\mathcal{A} \to Ab$

⇒ Freyd-Mitchell Embedding Theorem:

The main point is: by reduce the target category, to influence the ker/coker.

① Consider the failure of the Yoneda Embedding $h: \mathcal{A} \to Ab^{\mathcal{A}^{op}}$ to be right exact:

i.e. Given $0 \to A \xrightarrow{f} B \xrightarrow{g} C \to 0$ exact, define $W(M)$ in $Ab^{\mathcal{A}^{op}}$ by the exactness of:

$$0 \to \mathrm{Hom}_{\mathcal{A}}(M, A) \xrightarrow{f_*} \mathrm{Hom}_{\mathcal{A}}(M, B) \xrightarrow{g_*} \mathrm{Hom}_{\mathcal{A}}(M, C) \to \underbrace{W(M)}_{\mathrm{coker}(g_*)} \to 0 \quad \cdots (*)$$

In the sense that W respects morphism in $\mathcal{A} \Rightarrow W$ is a contravariant functor: $\mathcal{A} \to Ab$

$\Rightarrow$ exact sequence of functors: $0 \to h_A \to h_B \to h_C \to W \to 0$

One further states that W is a weakly effaceable functor. Denote the subcategory of $-$ as $\mathcal{W}$.

the quotient functor $R: Ab^{\mathcal{A}^{op}} \to \mathcal{L}$ s.t. $R(L) = L$ if $L \in \mathcal{L}$. $R(W) = 0$ if $W \in \mathcal{W}$

Rm. ① One proves that $\mathcal{W}$ is a localizing subcat. of $Ab^{\mathcal{A}^{op}}$, meaning that the quotient is well-defined.

② denote $\mathcal{T} = Ab^{\mathcal{A}^{op}}$, $i: \mathcal{L} \hookrightarrow \mathcal{T}$. then $\mathrm{Hom}_{\mathcal{L}}(RF), L) \simeq \mathrm{Hom}_{\mathcal{T}}(F, i(L))$

③ Composing $(*)$ w/ R, one obtain: $0 \to h_A \to h_B \to h_C \to 0$ □

④ $\mathcal{L}$ contains a faithful projective objective $P \Rightarrow R = \mathrm{Hom}_{\mathcal{L}}(P, P)$, then $\mathrm{Hom}_{\mathcal{L}}(P, A)$ is a right R-mod.

> 从弱可消去函子 W 到任意左正合函子 L 的自然变换，都只能是零。

我们把这个结论直接证明出来。设 $\tau : W \to L$ 是自然变换，任取 $M \in \mathcal{A}$ 和 $x \in W(M)$。由于 W 弱可消去，存在满态射 $p : P \twoheadrightarrow M$，使得 $W(p)(x) = 0$。

自然性给出交换图

$$\begin{array}{ccc} W(M) & \xrightarrow{\tau_M} & L(M) \\ {\scriptstyle W(p)}\downarrow & & \downarrow{\scriptstyle L(p)} \\ W(P) & \xrightarrow{\tau_P} & L(P). \end{array}$$

因此

$$L(p)\big(\tau_M(x)\big) = \tau_P\big(W(p)(x)\big) = 0.$$

另一方面，令 $K = \ker p$。短正合列 $0 \to K \to P \to M \to 0$ 被反变左正合函子 L 送成

$$0 \longrightarrow L(M) \xrightarrow{L(p)} L(P) \longrightarrow L(K).$$

所以 $L(p)$ 是单射。由此可知 $\tau_M(x) = 0$。由于 M, x 任意，整个自然变换 τ 都为零。

现在使用伴随：

$$\mathrm{Hom}_{\mathcal{C}}(R(W), L) \cong \mathrm{Hom}_{\mathcal{F}}(W, iL) = 0.$$

特别取 $L = R(W)$，得到 $\mathrm{Hom}_{\mathcal{C}}(R(W), R(W)) = 0$。这个群中本来应当包含恒等态射，因此 $\mathrm{id}_{R(W)} = 0$，从而 $R(W)$ 是零对象。

这个证明把两种性质的关系展示得很清楚：弱可消去性让一个元素在某个满态射之后变成零；左正合性则保证满态射诱导的反向映射是单射，因此左正合函子不能接收到这样的非零元素。

Details of Freyd-Mitchell Embedding Thm to be considered.

Derived Functors

δ-Functors

Def. Given 2 abelian categories $\mathcal{A}, \mathcal{B}$.

or w/ the convention that $T_n = 0,\ n<0$.

a homological δ-functor between $\mathcal{A}$ and $\mathcal{B}$ is a collection: $T_n: \mathcal{A}\to\mathcal{B}$ for $n\geq 0$ s.t.

For each SES $0\to A\to B\to C\to 0$ in $\mathcal{A}$

① we have the connecting morphisms: $\delta_n: T_n(C)\to T_{n-1}(A)$ that makes there a LES:

$$\cdots\to T_{n+1}(C)\xrightarrow{\delta} T_n(A)\to T_n(B)\to T_n(C)\xrightarrow{\delta} T_{n-1}(A)\to\cdots$$

in particular. T_0 is right exact. T^0 is left exact (for cohomology) to obtain a cutoff: $\to T_0(C)\to 0$

② δ commutes w/ morphism of SES,

i.e. for chain morphism: $\begin{array}{ccccc} 0\to A' & \to B' & \to C' & \to 0 \\ \downarrow & \downarrow & \downarrow & \\ 0\to A & \to B & \to C & \to 0 \end{array}$, we require: $\begin{array}{ccccccc} \cdots\to T_n(A') & \to T_n(B') & \to T_n(C') & \xrightarrow{\delta} & T_{n-1}(A') & \to\cdots \\ \downarrow & \downarrow & \downarrow & & \downarrow & \\ \cdots\to T_n(A) & \to T_n(B) & \to T_n(C) & \xrightarrow{\delta} & T_{n-1}(A) & \to\cdots \end{array}$

eg. (p-torsion) Given p an integer and consider Ab (the category of abelian groups). Apply Snake Lemma to ↳ an abelian category

$$\begin{array}{ccccc} 0\to A & \to B & \to C & \to 0 \\ p\downarrow & p\downarrow & p\downarrow & \\ 0\to A & \to B & \to C & \to 0 \end{array} \Rightarrow 0\to {}_pA\to {}_pB\to {}_pC\xrightarrow{\delta} A/pA\to B/pB\to C/pC\to 0.$$

$${}_pA := \{a\in A \mid pa=0\}.$$

then define: $T_0(A)=A/pA$, $T_1(A)={}_pA$ w/ δ the connecting morphism satisfies the definition of δ functor

---

Once we replace Ab w/ an arbitrary ring R, i.e. consider δ-functor from R-mod to Ab.

The similar construction: $S_0(M)=M/rM$, $S_1(M)={}_rM$ for $r\in R$ is still a δ-functor.

However. $T_n(M)=\mathrm{Tor}_n^R(R/r,M)$ is also a δ-functor, which might be distinct from S_n when $rR\neq 0$

⇒ It's possible to construct different δ-functor with the same starting point $T_0=S_0$

⇒ criterion needed to get a unique one.

10. 一个具体反例：$R=\mathbb{Z}/4,\ r=2$

令

$$R=\mathbb{Z}/4\mathbb{Z},\qquad r=2.$$

那么

$$K_2={}_2R=\{\bar{0},\bar{2}\}\neq 0,$$

所以 2 是零因子。

先取 $M=R$，朴素的 Snake Lemma δ-函子给出

$$T_1(R)=R[2]=\{\bar{0},\bar{2}\}\neq 0.$$

但是

$$\mathrm{Tor}_1^R(R/2R,R)=0,$$

因为 R 作为自身上的模是自由模，也就是平坦模。商公式也给出

$$\mathrm{Tor}_1^R(R/2R,R)\cong\frac{R[2]}{K_2R}=\frac{\{\bar{0},\bar{2}\}}{\{\bar{0},\bar{2}\}}=0.$$

所以这里确实有两个同调型 δ-函子具有相同的零次项

$$T_0(M)=M/2M,$$

但它们的一次项不同：

$$\text{朴素函子的一次项}=M[2],$$

$$\text{universal Tor 的一次项}=\mathrm{Tor}_1^R(R/2R,M).$$

甚至更高次数也可能非零。$R/2R$ 有周期自由分解

$$\cdots\xrightarrow{\times 2}R\xrightarrow{\times 2}R\xrightarrow{\times 2}R\longrightarrow R/2R\longrightarrow 0.$$

再与 $R/2R$ 张量时，所有乘 2 的映射都变成零，因此

$$\mathrm{Tor}_n^R(R/2R,R/2R)\cong R/2R$$

对所有 $n\geq 0$ 都成立。这说明一旦 r 是零因子，真实的导出信息可能一直延伸到任意高次，而不仅仅停留在 T_1。

11. 这为什么能解释“universal”的必要性？

Generalization 构造的朴素同调型 δ-函子是

$$S_0(M)=M/rM,\qquad S_1(M)=M[r],\qquad S_n(M)=0\ (n\geq 2).$$

由投射分解得到的 Tor 则给出

$$T_n(M)=\mathrm{Tor}_n^R(R/rR,M),$$

其中

$$T_0(M)\cong M/rM.$$

两者有同一个零次函子，但不一定具有相同的高次项。前一节 Remark 只说：

> 以给定 T_0 开始的 universal δ-函子至多有一个。

它并没有说所有以同一个 T_0 开始的普通 δ-函子都相同。现在这个例子正好展示了差别：

- Snake Lemma 很容易制造出一个普通 δ-函子；
- Tor 给出 universal δ-函子；
- 当 $rR=0$ 时，二者相同；
- 当 $rR\neq 0$ 时，二者通常不同，因而朴素的那个通常不是 universal。

所以这段 Generalization 并非只是把 p 换成 r 的无聊推广。它实际上是在为后面的问题做铺垫：

> “具有正确零次项的 δ-函子可能很多；为什么导出函子给出的那一个才是规范的？”

答案就是 universality。

Def: Given 2 δ-functors S, T.

a morphism $S \to T$ is a system of natural transformations that commutes w/ δ.

i.e. $$\begin{array}{ccccc} \cdots \to & S_n(C) & \xrightarrow{\delta} & S_{n-1}(A) & \to \\ & \downarrow & & \downarrow & \\ \cdots \to & T_n(C) & \xrightarrow{\delta} & T_{n-1}(A) & \to \end{array}$$ commutative ladder.

① A homological δ-functor T is universal if:

$$\begin{array}{ccccccc} S_0(A) & \to & S_0(B) & \to & S_0(C) & \to & 0 \\ \downarrow f_0 & & \downarrow f_0 & & \downarrow f_0 & & \\ T_0(A) & \to & T_0(B) & \to & T_0(C) & \to & 0 \end{array}$$

Given another δ-functor S and a natural transformation $f_0: S_0 \to T_0$, i.e.

then $\exists!\ \{f_n: S_n \to T_n\}$ of δ-functors that extends f_0 $\Rightarrow$ left derived for right exact

Then T_n are called the left satellite functors of T_0

② A cohomological δ-functor T is universal if:

$$\begin{array}{ccccccc} 0 \to & T^0(A) & \to & T^0(B) & \to & T^0(C) & \\ & \downarrow f^0 & & \downarrow f^0 & & \downarrow f^0 & \\ 0 \to & S^0(A) & \to & S^0(B) & \to & S^0(C) & \end{array}$$

Given another δ-functor S and a natural transformation $f^0: T^0 \to S^0$, i.e.

then $\exists!\ \{f^n: T^n \to S^n\}$ of δ-functors that extends f^0 $\Rightarrow$ right derived for left exact.

Then T^n are called the right satellite functors of T^0

e.g. if $F: \mathcal{A} \to \mathcal{B}$ is an exact functor.

then $T_0 = F$, $T_n = 0$ defines a universal δ-functor of both homology & cohomology type.

pf: for $n \neq 0$, the only natural transformation $f_n: S_n \to \overset{0}{T_n}$ (resp. $f^n: \overset{0}{T^n} \to S^n$) is zero morphism

— — — —

Further, if the universal δ-functor exists, then it's unique.

One then steps forward to the question of how to construct the higher order T_n from given T_0.

Projective Resolution.

Def. An object P in an abelian category $\mathcal{A}$ is projective if it satisfies the universal lifting property:

Given a surjection $g: B \to C$, a map $\gamma: P \to C$ $\quad \exists \beta: P \to B$

$\Rightarrow \exists$ at least one map $\beta: P \to B$ that commutes: $\quad \begin{array}{ccc} & & P \\ & \swarrow^{\beta} & \downarrow \gamma \\ B & \xrightarrow{g} & C \end{array}$

Rm ① the question is that though we could always find a lift b_p s.t. $g(b_p) = \gamma(p)$ $\forall p \in P$ however, we shall verify it's a morphism: $b_{p+q} = b_p + b_q$.

② the lift could not be unique: $g\beta_1 = \gamma = g\beta_2 \Rightarrow \beta_1 - \beta_2 \in \ker g$ is controlled by: $P \to \ker g$.

③ if $\{P_\alpha\}$ is a family of projectives $\Rightarrow$ then the coproduct $\coprod P_\alpha$ is also projective. (might be infinite)

Then we steps forward to certain equivalent definition, or properties of projective objective.

① P projective $\Leftrightarrow$ $\mathrm{Hom}_R(P,-)$ exact.

For SES $0\to A\to B\xrightarrow{g} C\to 0$, one have $0\to \mathrm{Hom}_R(P,A)\to \mathrm{Hom}_R(P,B)\xrightarrow{g_*}\mathrm{Hom}_R(P,C)$

$\mathrm{Hom}_R(P,-)$ exact $\Leftrightarrow$ right exact $\Leftrightarrow$ Given surjection $g:B\to C$, map $\gamma:P\to C$, $\exists\,\beta:P\to B$ s.t. $\gamma = g_*(\beta)=g\circ\beta$

② P projective $\Leftrightarrow$ every surjection $E\to P$ splits. i.e. every $0\to K\to E\xrightarrow{g} P\to 0$ splits.

I. Given P projective. Use the lifting property w/ $\gamma=\mathrm{id}_P$: $\exists s$ with $g\circ s=\mathrm{id}_P$ $\Rightarrow \exists\, s:P\to E$ s.t. $g\circ s=\mathrm{id}_P$

Rm: Given s s.t. $g\circ s=\mathrm{id}_P$, s is called a section of g, g is called a splitting surjection.

- One immediately has that: s is injective. since if $s(p)=0\Rightarrow p=g\circ s(p)=0$
- Specifically: $\forall x\in E$. we could decompose it: $x=\underbrace{(x-sg(x))}_{\in\ker g=K}+\underbrace{sg(x)}_{\in \mathrm{Im}(sg)=\mathrm{Im}(s)=s(P)}$

if $s(p)\in\ker g \Rightarrow g\,s(p)=p=0$ $\Rightarrow$ trivial intersection,

$\Rightarrow E=i(K)\oplus s(P)\simeq K\oplus P$ ($i:K\hookrightarrow E$ the inclusion. since s is injection $\Rightarrow s(P)\simeq P$)

II. Backwards. assume all $E\to P$ splits

Given general surjection $g:B\twoheadrightarrow C$, $\gamma:P\to C$, construct the pullback of γ:

$P\times_C B=\{(p,b)\in P\oplus B:\gamma(p)=g(b)\}$ w/ a surjective projection $\pi_P: P\times_C B\to P$

all $E\to P$ splits $\Rightarrow \exists s:P\to P\times_C B$ s.t. $\pi_P\circ s=\mathrm{id}_P$. Compose s w/ the projection $\pi_B: P\times_C B\to B$

$\Rightarrow \beta:=\pi_B\circ s$ is as required. since $g\circ\beta(p)=g\circ\pi_B\circ s(p)=g\circ\pi_B((p,b_p))=g(b_p)=\gamma(p)$ □

(b_p s.t. $g(b_p)=\gamma(p)$)

Criteria: To prove P is not projective, we only need to find a surjection $E\twoheadrightarrow P$ that doesn't split.

e.g.: $\mathbb{Z}\to\mathbb{Z}/n\mathbb{Z}$ doesn't split, therefore $\mathbb{Z}/n\mathbb{Z}$ is not a projective $\mathbb{Z}$-module,

(but is a projective $\mathbb{Z}/n\mathbb{Z}$ module)

③ P projective $\Leftrightarrow$ P is a summand (component of direct sum) of a free module E

First: every free module is projective. totally different from the concept of submodule.

Let $F=\bigoplus_{i\in I} R e_i$ be the free left module for given basis $\{e_i\}_{i\in I}$

Given surjection $g:B\twoheadrightarrow C$, $\gamma:F\to C$. for every basis e_i, choose $b_i\in B$ s.t. $g(b_i)=\gamma(e_i)$

Since F is free $\Rightarrow \beta(e_i)=b_i$ would be a homomorphism $\beta:F\to B$

Rm: definition of free module: (ideal: no extra constraints on the basis)

① a module w/ a basis s.t. every element could be written as a finite linear combination of the basis.

e.g. treat $\mathbb{Z}/n\mathbb{Z}$ as a $\mathbb{Z}$-module. it's finitely generated by $\bar{1}$, but $n\bar{1}=0$, $n\neq 0$ in $\mathbb{Z}$

$\Rightarrow \{\bar{1}\}$ is linearly dependent. not a free $\mathbb{Z}$-module.

However, treat $\mathbb{Z}/n\mathbb{Z}$ itself as a $\mathbb{Z}/n\mathbb{Z}$ module, it's both finitely generated and free, since $\bar{n}=\bar{0}$ in $\mathbb{Z}/n\mathbb{Z}$

② $M \simeq \bigoplus_{i\in I} R$. (when I is infinite, we require element to have finite none-zero component.

③ universal property: choose a basis as $\{e_i\}_{i\in I}$, for arbitrary R-module N w/ map $f_i: e_i \mapsto n_i$

$\Rightarrow \exists!$ R-linear map $f: F \to N$ $f(\sum_i r_i e_i) = \sum_i r_i n_i$

Further, every R-module is a quotient of certain free R-module:

let $\{m_i\}_{i\in I}$ be the generating set of R-module M. construct $F = \bigoplus_{i\in I} R m_i \Rightarrow$ surjection $F \twoheadrightarrow M \Rightarrow M = F/\ker(F \twoheadrightarrow M)$

---

① Assume P projective use the above,

$\Rightarrow$ construct $Fr(P) = \bigoplus_{p\in P} R[p]$, where $[p]$ is the formal basis w.r.t. P

$\Rightarrow Fr(P)$ is the free-module using the underlying set structure of P

Then, naturally we have a surjection $\pi: Fr(P) \twoheadrightarrow P$, $[P] \mapsto P$

Since P projective, we have a lift $i: P \to Fr(P)$ s.t. $\pi i = id \Rightarrow \pi$ split $\Rightarrow Fr(P) = P \oplus \ker\pi$

② Conversely, assume $F \simeq P \oplus Q$, F free. Given $g: B \to C$, $\gamma: P \to C$

Denote $i: P \hookrightarrow F$, $\pi: F \to P$, then we construct: $\gamma\circ\pi: F \to C$.

since F free $\Rightarrow$ lift to $\alpha: F \to B$ s.t. $g\alpha = \gamma\circ\pi$

Then we construct: $\beta = \alpha i \Rightarrow g\beta = g\alpha i = \gamma\pi i = \gamma \Rightarrow P$ projective.

e.g. the category $\mathcal{A}$ of finite abelian groups has no nonezero projective objects.

pf: use the structure thm for finite abel group:

$P \simeq \mathbb{Z}/p^m\mathbb{Z} \oplus Q$ for p prime, $m\geq 1$ and Q another abelian group

then construct $B = \mathbb{Z}/p^{m+1}\mathbb{Z} \oplus Q$ w/ surjection $g: B \to P$, $(\bar{x}, y) \mapsto (\bar{x} \bmod p^m, y)$

if P projective $\Rightarrow \exists$ section $s: P \to B$ s.t. $gs = id_P$

particularly, consider $s(\bar{1}, 0) = (\bar{a}, 0)$ w/ $a \equiv 1 \pmod{p^m}$

since $p^m(\bar{1}, 0) = 0 \Rightarrow p^m s(\bar{1}, 0) = 0 \Rightarrow p^m\bar{a} = 0 \pmod{p^{m+1}}$

However, since $a \equiv 1 \pmod{p^m} \Rightarrow a = 1 + tp^m \Rightarrow p^m\bar{a} = p^m + tp^{2m} \equiv p^m \pmod{p^{m+1}} \neq 0$ □

$\Rightarrow P$ is not projective.

Def: we say that $\mathcal{A}$ has enough projectives if for $\forall A \in \mathcal{A}$, there is a surjection $P \to A$ w/ P projective

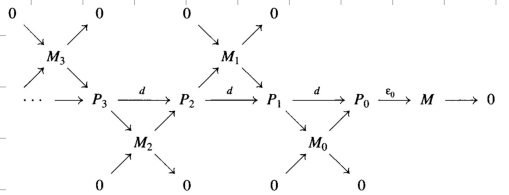
Def: Let $M \in \mathcal{A}$, a left resolution of M is a complex $P_\bullet$ w $P_i = 0$, $i < 0$, together w/ $\epsilon: P_0 \to M$ s.t.

$\cdots \xrightarrow{d} P_2 \xrightarrow{d} P_1 \xrightarrow{d} P_0 \xrightarrow{\epsilon} M \to 0$ is exact.

It's a projective resolution if each P_i is projective.

Lem - If an abelian category $\mathcal{A}$ has enough projectives, then $\forall M \in \mathcal{A}$, admits a projective resolution.

Particularly, every R-module admits a projective resolution.



Now we state the theorem that states the uniqueness of projective resolution, but in the sense of lifting property:

Comparison Theorem 2.2.6 *Let $P_\bullet \xrightarrow{\epsilon} M$ be a projective resolution of M and $f': M \to N$ a map in $\mathcal{A}$. Then for every resolution $Q_\bullet \xrightarrow{\eta} N$ of N there is a chain map $f: P_\bullet \to Q_\bullet$ lifting f' in the sense that $\eta \circ f_0 = f' \circ \epsilon$. The chain map f is unique up to chain homotopy equivalence.*

too strong → only requires:

Given a chain complex $\cdots \to P_1 \to P_0 \to M \to 0$ w/ P_i projective (No need to be exact)

No need to be projective.

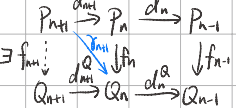
$$\begin{array}{ccccccccc} \cdots & \longrightarrow & P_2 & \longrightarrow & P_1 & \longrightarrow & P_0 & \xrightarrow{\epsilon} & M \longrightarrow 0 \\ & & \exists\downarrow & & \exists\downarrow & & \exists\downarrow & & \downarrow f' \\ \cdots & \longrightarrow & Q_2 & \longrightarrow & Q_1 & \longrightarrow & Q_0 & \xrightarrow{\eta} & N \longrightarrow 0 \end{array}$$

Existness: prove through induction.

① $n=0$:

$$\begin{array}{ccc} P_0 & \xrightarrow{\epsilon} & M \\ & & \downarrow f' \\ Q_0 & \xrightarrow{\eta} & N \to 0 \end{array} \quad \Rightarrow \quad \begin{array}{ccc} P_0 & & \\ & \searrow^{f'\circ\epsilon} & \\ Q_0 & \xrightarrow{\eta} & N \to 0 \end{array} \quad \Rightarrow \quad \begin{array}{c} P_0 \\ \downarrow f_0 \\ Q_0 \end{array}$$

② Given lifted $f_0, \cdots, f_n$, $n \geq 0$. now we construct f_{n+1} (one may treat f' as f_{-1})



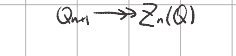
One expect to construct f_{n+1} s.t. $d^Q_{n+1} f_{n+1} = f_n d^P_{n+1} =: \gamma_{n+1}$

But d^Q_{n+1} might not be surjective

⇒ consider the lift restricted to $\mathrm{Im}(d^Q_{n+1}) = \ker(d^Q_n) = Z_n(Q)$

what left is to prove that $\mathrm{Im}(\gamma_{n+1}) \subset \mathrm{Im}(d^Q_{n+1}) = \ker(d^Q_n) = Z_n(Q)$

$\Rightarrow d^Q_n \gamma_{n+1} = d_n f_n d^P_{n+1} = f_{n-1} d^P_n d^P_{n+1} = 0$ □



Uniqueness:

Given 2 lift f, g w.r.t. f', denote $h = f - g$.

we shall construct a family $\{S_n : P_n \to Q_{n+1}\}$ s.t. $h_n = d^Q_{n+1} S_n + S_{n-1} d^P_n$

① $n=0$

$P_0 \xrightarrow{\epsilon} M$, $P_0 \xrightarrow{h_0} Q_0$, $Q_1 \xrightarrow{d} Q_0 \xrightarrow{\eta} M$

Since $\eta h_0 = \eta(f_0 - g_0) = 0\epsilon = 0 \Rightarrow \mathrm{im}(h_0) \subseteq \ker(\eta) = Z_0(Q)$

Through exactness of Q: $Z_0(Q) = \ker(\eta) = \mathrm{im}(d: Q_1 \to Q_0)$

$\Rightarrow$ $S_0 : P_0 \to Q_1$, $h_0 : P_0 \to Z_0(Q)$, $Q_1 \to Z_0(Q)$

$\Rightarrow h_0 = d^Q_1 \circ S_0 + S_{-1} \circ \epsilon$ (define $S_{-1} : M \to Q_0 = 0$)

② Assume $S_0, \cdots, S_{n-1}$ is constructed, s.t. $h_k = d^Q_{k+1} \circ S_k + S_{k-1} \circ d^P_k$, $k < n$

now we expect to construct S_n s.t. $h_n = d^Q_{n+1} \circ S_n + S_{n-1} \circ d^P_n$

$\Leftrightarrow d^Q_{n+1} \circ S_n = h_n - S_{n-1} \circ d^P_n =: \gamma_n$, the same lifting problem

Since $d^Q_n \gamma_n = d^Q_n (h_n - S_{n-1} \circ d^P_n) = d^Q_n h_n - (h_{n-1} - S_{n-2} \circ d^P_{n-1}) \circ d^P_n = d^Q_n h_n - h_{n-1} d^P_n = 0$

Similarly: $\exists S_n : P_n \to Q_{n+1}$, $\gamma_n : P_n \to Z_n(Q)$, $Q_{n+1} \xrightarrow{d^Q_{n+1}} Z_n(Q) \xrightarrow{d^Q_n} 0$ □.

Before moving on, we shall verify the uniqueness of projective resolution.

let $P_\cdot \to M$, $Q_\cdot \to M$ both be projective resolution.

$P_\cdot \to M$, $f \downarrow$, $Q_\cdot \to M$ and $Q_\cdot \to M$, $g \downarrow$, $P_\cdot \to M$

then, both $g \circ f$ and id_P is the lift of $P_\cdot \to M$, $P_\cdot \to M$

$\Rightarrow g \circ f \simeq id_P$. similarly $f \circ g \simeq id_Q$

$\Rightarrow$ these two projective resolutions are chain homotopy equivalent to each other.

There is another way to construct projective resolutions, called the Horseshoe Lemma

Given a SES $0 \to A' \xrightarrow{i} A \xrightarrow{\pi} A'' \to 0$ w/ projective resolutions $P'_\cdot \to A'$, $P''_\cdot \to A''$

then: $P_n = P'_n \oplus P''_n$ forms a projective resolution $P_\cdot \to A$

pf: ① lift $\epsilon'' : P'' \to A''$ to: $\beta : P'' \to A$

$\exists \beta : P''_0 \to A$, $P''_0 \xrightarrow{\epsilon''} A'' \to 0$

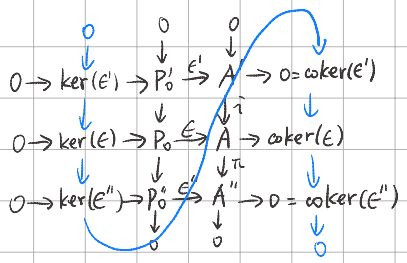
② construct $\epsilon : P_0 = P'_0 \oplus P''_0 \to A$, $(p', p'') \mapsto i\epsilon'(p') + \beta(p'')$ that the diagram commutes

$0 \to P'_0 \xrightarrow{\epsilon'} A' \to 0$

$P_0 \xrightarrow{\epsilon} A$

$P''_0 \xrightarrow{\epsilon''} A'' \to 0$

③ use snake lemma.



$\Rightarrow$ so $0 \to \ker(\epsilon') \to \ker(\epsilon) \to \ker(\epsilon'') \to 0$ is exact

$0 \to \operatorname{coker}(\epsilon) \to 0 \Rightarrow \operatorname{coker}(\epsilon) = 0$

instead, the surjectivity could also be verified through the exactness of $0 \to A' \to A \to A'' \to 0$

pf: Given a, first choose $x'' \in P_0''$ s.t. $\epsilon''(x'') = \pi(a)$

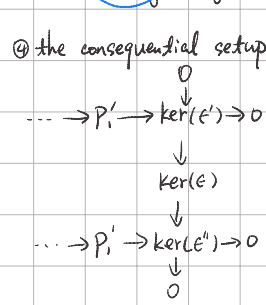
then: $a = (a - \beta(x'')) + \beta(x'')$

since $\pi(a - \beta(x'')) = \underbrace{\epsilon''(x'') - \pi \circ \beta}_{\epsilon''}(x'') = 0$

$\Rightarrow a - \beta(x'') \in \ker \pi = \operatorname{im} i$

$\Rightarrow \exists x' \in P_0'$ s.t. $a - \beta(x'') = i \circ \epsilon'(x')$

④ the consequential setup:



Rmk: ① the differential of the resolution $P_\cdot \to A$ has the general form:

$$d_n = \begin{pmatrix} d_n' & \alpha_n \\ 0 & d_n'' \end{pmatrix}, \quad \alpha_n : P_n'' \to P_{n-1}', \text{ which encodes how } A'' \text{ is extended by } A' \text{ to obtain } A$$

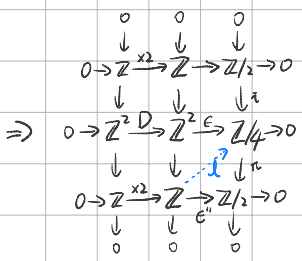
s.t. $d_n' \alpha_{n-1} + \alpha_n d_{n-1}'' = 0$

② the corollary is: Given $A' \to A \to A''$, through the comparison thm we may obtain $P_\cdot' \to P_\cdot \to P_\cdot''$.

further the Horseshoe lemma state that projective resolution preserves exactness (and is split)

e.g. for $0 \to \mathbb{Z}/2 \xrightarrow{i} \mathbb{Z}/4 \xrightarrow{\pi} \mathbb{Z}/2 \to 0$, $i([a]_2) = [2a]_4$, $\pi([a]_4) = [a]_2$

use the resolution: $0 \to \mathbb{Z} \xrightarrow{\times 2} \mathbb{Z} \to \mathbb{Z}/2 \to 0$



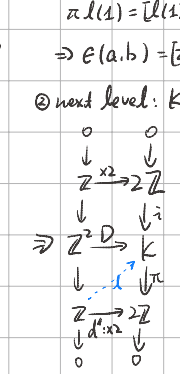
① lift ϵ'' to l s.t. $\pi l = \epsilon''$

since $\mathbb{Z}$ generated by 1, we only need to determine $l(1)$

$\pi l(1) = [l(1)]_2 = \epsilon''(1) = [1]_2 \Rightarrow l(1) = 1$.

$\Rightarrow \epsilon(a,b) = [2a+b]_4$ (for $\mathbb{Z}_2 \oplus \mathbb{Z}_2$, $\epsilon(a,b) = (a,b)$)

② next level: $K := \ker \epsilon = \{(a,b) \in \mathbb{Z}^2 \mid 2a+b \equiv 0 \pmod 4\}$



w/ $i(x) = (x,0)$, $\pi(x,y) = y$ is exact

since $(a,b) \in K \Rightarrow b$ is even $\Rightarrow \pi$ surjective

if $(a,b) \in \ker \pi \Rightarrow b = 0 \Rightarrow 2a \equiv 0 \pmod 4 \Rightarrow a \in 2\mathbb{Z} = \operatorname{im}(i)$

then lift d'' to l s.t. $\pi l = d''$

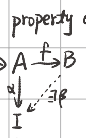
$\Rightarrow l(1) = (c, 2) \in K \Rightarrow 2c + 2 \equiv 0 \pmod 4 \Rightarrow c = 1$

$\Rightarrow D = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$ (for $\mathbb{Z}_2 \oplus \mathbb{Z}_2$, $D = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$)

Injective Resolution.

Def: Dually: an object I in an abelian category $\mathcal{A}$ has the lifting property of extending the domain, i.e.

Given an injection $f: A \to B$ and a map $\alpha: A \to I$ then $\Rightarrow$ exists at least one $\beta: B \to I$ s.t. $\alpha = \beta \circ f$



Rm① also. if $\{I_\alpha\}$ is a family of injectives $\Rightarrow$ so if the product $\prod I_\alpha$ (might be infinite)

② Injective module is invented by R. Baer in 1940.

long before 1956 when projective module is invented by Cartan & Eilenberg.

$\Rightarrow$ the dual criterions:

I injective ① $\Leftrightarrow$ $\mathrm{Hom}_{\mathcal{A}}(-, I)$ exact ② $\Leftrightarrow$ every injection $I \hookrightarrow Y$ splits (i.e. $\exists$ left inverse) ③

prove: ①$\Leftrightarrow$② is trivial.

①$\Rightarrow$③. $I \xhookrightarrow{i} Y$, $\mathrm{id}_I: I \to I$, $\exists r: Y \to I$ $\Rightarrow \exists r: Y \to I$ s.t. $ri = \mathrm{id}_I$. $\Rightarrow Y \simeq I \oplus \ker r$

③$\Rightarrow$①: Given $A \hookrightarrow B$, $f: A \to I$, construct $Y = (I \oplus B) / \{(f(a), -a) \mid a \in A\}$.

then $\Rightarrow$ $i: I \hookrightarrow Y$. $x \mapsto [(x, 0)]$ is injective. $\Rightarrow$ splits: $\exists r: Y \to I$ s.t. $ri = \mathrm{id}_I$

extend: def $\tilde{f}: B \to I$, $b \mapsto r([(0, b)])$, then: $\tilde{f}|_A: a \mapsto r([(0, a)]) = r([(f(a), 0)]) = f(a)$

Baer's Criterion: a right R-module E is injective $\Leftrightarrow$ $\forall$ right ideal J of R, every map $J \to E$ can be extend to $R \to E$

①$\Rightarrow$": by definition

②"$\Leftarrow$": suppose given an R-module $B \supseteq A$ a submodule and $\alpha: A \to E$

denote $\mathcal{E}$ as the poset of all extensions $\alpha': A' \to E$ w/ partial order $\alpha' \le \alpha''$ if α'' extends α'.

Since it admits a upper bound $A_\infty = \bigcup_\lambda A_\lambda$ s.t. $\alpha' \le \alpha_\infty$. Zorn Lemma $\Rightarrow \exists$ maximal extension $\alpha': A' \to E$

Assume: $A' \subsetneq B$ $\Rightarrow$ choose $b \in B \setminus A'$, then try to analyse the obstructions of extension to: $A' + bR$

if we want to define extension α'': $A' + bR \to E$, then: $\alpha''(a + br) = \alpha'(a) + \alpha''(b)r$ by R-linearity

the question of well-defineness occurs when: $br \in A'$, then $\alpha''(b)r = \alpha'(b)r$

Thus we test the right ideal: $J = \{r \in R \mid br \in A'\}$

then we have: $J \xrightarrow{b} A' \xrightarrow{\alpha'} E$, which by assumption extends to a map $f: R \to E$

then we define: $\alpha''(a + br) = \alpha'(a) + f(r)$, which is well-defined, therefore $\Rightarrow A' = B$, I is injective.

Corollary: an abelian group E is a $\mathbb{Z}$-module, then it's injective $\Leftrightarrow$ it's divisible, i.e. $\forall n \ge 1, x \in E, \exists y \in E$ s.t. $ny = x$

More generally: if R is a PID, then an R-module is injective $\Leftrightarrow$ is divisible.

e.g. $\mathbb{Z}$ is not injective, $\mathbb{Q}$, $\mathbb{Z}_{p^\infty} = \mathbb{Z}[\frac{1}{p}]/\mathbb{Z} = \varinjlim_k \mathbb{Z}/p^k\mathbb{Z}$ are divisible and therefore injective.

↑ Prüfer p-group

where we use the inclusion: $p^{-1}\mathbb{Z}/\mathbb{Z} \hookrightarrow p^{-2}\mathbb{Z}/\mathbb{Z} \hookrightarrow \cdots$, where $p^{-k}\mathbb{Z}/\mathbb{Z} = \{0, \frac{1}{p^k}, \cdots, \frac{p^k-1}{p^k}\}$

In fact, every injective abelian group is a direct sum of $\mathbb{Q}$ and $\mathbb{Z}_{p^\infty}$

i.e. $D \simeq \mathbb{Q}^{(K_0)} \oplus \bigoplus_{p \text{ prime}} \mathbb{Z}_{p^\infty}^{(K_p)}$, which is achieved by spliting the torsion part that $\simeq \mathbb{Z}_{p^\infty}$

particularly: $\mathbb{Q}/\mathbb{Z} \simeq \bigoplus_{p \text{ prime}} \mathbb{Z}_{p^\infty}$ (By Chinese Reminder Theorem)

$\Rightarrow$ One could then prove that Ab (the category of abelian groups) has enough injectives

pf: Denote $S = \mathrm{Hom}_{Ab}(A, \mathbb{Q}/\mathbb{Z})$, and define $I(A) = \prod_{f \in S} \mathbb{Q}/\mathbb{Z}$. which is injective

then we define a canonical map $e_A : A \to I(A)$, $a \mapsto (f(a))_{f \in S}$

to show e_A is injective, we shall prove if $a \neq 0$, then $\exists f : A \to \mathbb{Q}/\mathbb{Z}$ s.t. $f(a) \neq 0$

i.e. $A = 0 \Leftrightarrow \mathrm{Hom}_{Ab}(A, \mathbb{Q}/\mathbb{Z}) = 0$

we first consider the subgroup $\langle a \rangle$ generated by a

① if $\langle a \rangle$ is of finite order $n \geq 2$, then define: $\psi : \langle a \rangle \to \mathbb{Q}/\mathbb{Z}$, $ka \mapsto \frac{k}{n} + \mathbb{Z}$

② if $\langle a \rangle$ is of infinite order, then $\langle a \rangle \simeq \mathbb{Z}$, define: $\psi : \mathbb{Z} \to \mathbb{Q}/\mathbb{Z}$, $k \mapsto \frac{k}{2} + \mathbb{Z}$

the use the injectivity of $\mathbb{Q}/\mathbb{Z}$ to lift to: $\tilde{\psi} : A \to \mathbb{Q}/\mathbb{Z}$, w/ $\tilde{\psi}(a) \neq 0$ □

We now states an abstract but meaningful lemmma:

Lem. $\forall$ right R-module M, $\tau : \mathrm{Hom}_{Ab}(M, A) \to \mathrm{Hom}_{\text{mod-}R}(M, \mathrm{Hom}_{Ab}(R, A))$, $\tau f : (m, r) \mapsto f(mr)$ is an isomorphism.

(right action is: Given $\varphi : R \to A$, $k \in R$, $(\varphi \cdot k)(r) = \varphi(kr)$ to satisfy the associativity.)

pf: Backwards Given $g : M \to \mathrm{Hom}_{Ab}(R, A)$, R linearity shows that: $g(m)(r) = g(mr)(1)$

we define the inverse map μ of τ s.t.: $\mu g : m \mapsto g(m)(1)$

then: $\tau(\mu g)(m, r) = \mu g(mr) = g(mr)(1) = g(m)(r)$

$\mu(\tau f)(m) = \tau f(m)(1) = f(m \cdot 1) = f(m)$ □

In other word, consider the forgetful functor $U : \text{mod-}R \to Ab$

$\Rightarrow \mathrm{Hom}_{Ab}(U(M), A) \cong \mathrm{Hom}_{\text{mod-}R}(M, \mathrm{Hom}_{Ab}(R, A)) \Rightarrow U \dashv \mathrm{Hom}_{Ab}(R, -)$

Definition 2.3.9 A pair of functors $L : \mathcal{A} \to \mathcal{B}$ and $R : \mathcal{B} \to \mathcal{A}$ are *adjoint* if there is a natural bijection for all A in $\mathcal{A}$ and B in $\mathcal{B}$:

$$\tau = \tau_{AB} : \mathrm{Hom}_{\mathcal{B}}(L(A), B) \xrightarrow{\cong} \mathrm{Hom}_{\mathcal{A}}(A, R(B)).$$

Here "natural" means that for all $f : A \to A'$ in $\mathcal{A}$ and $g : B \to B'$ in $\mathcal{B}$ the following diagram commutes:

$$\begin{array}{ccccc} \mathrm{Hom}_{\mathcal{B}}(L(A'), B) & \xrightarrow{Lf^*} & \mathrm{Hom}_{\mathcal{B}}(L(A), B) & \xrightarrow{g_*} & \mathrm{Hom}_{\mathcal{B}}(L(A), B') \\ \downarrow\tau & & \downarrow\tau & & \downarrow\tau \\ \mathrm{Hom}_{\mathcal{A}}(A', R(B)) & \xrightarrow{f^*} & \mathrm{Hom}_{\mathcal{A}}(A, R(B)) & \xrightarrow{Rg_*} & \mathrm{Hom}_{\mathcal{A}}(A, R(B')). \end{array}$$

We call L the *left adjoint* and R the *right adjoint* of this pair. The above lemma states that the forgetful functor from **mod**–R to **Ab** has $\mathrm{Hom}_{\mathbf{Ab}}(R, -)$ as its right adjoint.

Prop. Given $L: A \rightleftarrows B : R$, i.e. $L \dashv R$.

① if R additive, L (left) exact, then $\Rightarrow$ R preserves injectives

Dually ② if L additive, R (right) exact, then $\Rightarrow$ L preserves projectives

pf: ① we shall prove that $\mathrm{Hom}_A(-, R(I))$ is exact, the only need is to prove it's right exact (preserving onto)

Given an injection $f: A \hookrightarrow A'$, L (left) exact $\Rightarrow Lf: L(A) \hookrightarrow L(A')$ is injective.

for injectives I in $B \Rightarrow \mathrm{Hom}_B(L(A'), I) \to \mathrm{Hom}_B(L(A), I) \to 0$

$\tau_{A'I} \downarrow \cong$ $\tau_{AI} \downarrow \cong$

$\mathrm{Hom}_A(A', R(I)) \to \mathrm{Hom}_A(A, R(I))$

Coro: ① if $I \in Ab$ is injective, then $\Rightarrow \mathrm{Hom}_{Ab}(R, I)$ is an injective R module.

Left Derived Functors.

For right exact functor $F: A \to B$, A, B abelian category. If A has enough projectives

$\Rightarrow$ construct the left derived functor $L_iF (i \geq 0)$ as:

Given $A \in Ob(A)$, choose (once and for all) a projective resolution $P_\cdot \to A$,

define: $L_iF(A) := H_i(F(P))$

Rm. ① since F is right exact $\Rightarrow F(P_1) \xrightarrow{d_1} F(P_0) \to F(A) \to 0$ is exact

$\Rightarrow L_0F(A) = F(P_0)/\mathrm{im}\, d_1 \simeq F(A)$

② **Lemma 2.4.1** *The objects $L_iF(A)$ of $\mathcal{B}$ are well defined up to natural isomorphism. That is, if $Q \to A$ is a second projective resolution, then there is a canonical isomorphism:*

$$L_iF(A) = H_i(F(P)) \xrightarrow{\cong} H_i(F(Q)).$$

In particular, a different choice of the projective resolutions would yield new functors $\hat{L}_iF$, which are naturally isomorphic to the functors L_iF.

already proven.

② Coro. of ↑ $\Rightarrow$ if A is projective, then $L_iF(A) = 0$, $i > 0$

it's just a complex, not restrict to be exact.

pf. use the projective resolution: $0 \to A \xrightarrow[\sim]{id} A \to 0 \Rightarrow F(P) = 0 \to F(A) \to 0$

Def. F-Acyclic Objects are Q's that $L_iF(Q) = 0$, $\forall i \neq 0$ (clearly: projective $\overset{*}{\Rightarrow}$ F-acyclic)

(will be used for calculations of left derived functor $\Rightarrow$ F-acyclic resolution)

③ By the comparison thm. Given $f: A' \to A$ in $A \Rightarrow \exists$ a natural map:

$L_iF(f): L_iF(A') \to L_iF(A) \Rightarrow L_0F(f) = F(f)$

④ The above $\Rightarrow$ each L_iF is an additive functor: $A \to B$

Exercise 2.4.2 (Preserving derived functors) If $U: \mathcal{B} \to \mathcal{C}$ is an exact functor, show that

$$U(L_i F) \cong L_i(UF).$$

Forgetful functors such as **mod**–$R \to$ **Ab** are often exact, and it is often easier to compute the derived functors of UF due to the absence of cluttering restrictions.

consider SES: $0 \to Z_i \to C_i \to B_{i-1} \to 0$

$\Rightarrow 0 \to U(Z_i) \to U(C_i) \to U(B_{i-1}) \to 0$

$\Rightarrow B_i(U(C_\cdot)) = U(B_i),\ Z_i(U(C_\cdot)) = U(Z_i)$

$\to H_i(U(C_\cdot)) = U(H_i)$

We would show that: $L_*\bar{F}$ form a universal homological δ-functor.

(a) It's a δ-functor.

Given SES $0 \to A' \to A \to A'' \to 0$ in $\mathcal{A}$. from the Horseshoe Lemma, consider $P'_\cdot \to A'$. $P''_\cdot \to A''$

$\to \exists P_\cdot \to A$ s.t. $0 \to P'_n \to P_n \to P''_n \to 0$ is split exact.

$\Rightarrow 0 \to F(P'_n) \to F(P_n) \to F(P''_n) \to 0$ is split exact. $\Rightarrow 0 \to F(P'_\cdot) \to F(P_\cdot) \to F(P''_\cdot) \to 0$ is exact.

$\Rightarrow$ connecting homomorphism $\delta: H_n(F(P''_\cdot)) \to H_{n-1}(F(P'_\cdot))$

Now we verify that δ commutes w/ morphisms of SES

Given: $\begin{array}{ccccccc} 0 & \to & A' & \xrightarrow{i_A} & A & \xrightarrow{\pi_A} & A'' \to 0 \\ & & \downarrow f' & & \downarrow f & & \downarrow f'' \\ 0 & \to & B' & \xrightarrow{i_B} & B & \xrightarrow{\pi_B} & B'' \to 0 \end{array}$, we shall prove: $L_{n-1}F(f')\delta_n^A = \delta_n^B \circ L_n F(f'')$

Given: $\epsilon': P'_\cdot \to A'$. $\epsilon'': P''_\cdot \to A''$, $\eta': Q'_\cdot \to B'$, $\eta'': Q''_\cdot \to B''$

use the horseshoe lemma and the comparison thm $\Rightarrow$ natural lift f', f'' to F', F''

$\Rightarrow$ we shall construct F, as a lift of f that the diagram commutes.

$\begin{array}{ccccccc} 0 & \to & P' & \xrightarrow{i_A} & P & \xrightarrow{\pi_A} & P'' \to 0 \\ & & \downarrow F' & & \vdots F & & \downarrow F'' \\ 0 & \to & Q' & \xrightarrow{i_B} & Q & \xrightarrow{\pi_B} & Q'' \to 0 \end{array}$

Now we construct $\gamma_n: P''_n \to Q'_n$ (not required to commute w/ the differential, i.e. not chain map)

s.t. $F_n = \begin{bmatrix} F'_n & \gamma_n \\ & F''_n \end{bmatrix} : \begin{matrix} P'_n \\ \oplus \\ P''_n \end{matrix} \longrightarrow \begin{matrix} Q'_n \\ \oplus \\ Q''_n \end{matrix}$ This ensures that the diagram commutes. we only need to verify F is a chain map.

① At $n=0$, it's to verify F_0 is a lift of f

$\begin{array}{ccc} P'_0 \oplus P''_0 & \xrightarrow{\epsilon} & A \to 0 \\ \downarrow F_0 & & \downarrow f \\ Q'_0 \oplus Q''_0 & \xrightarrow{\eta} & B \to 0 \end{array}$ $\Rightarrow \eta F_0 - f\epsilon = 0 : P'_0 \oplus P''_0 \to B$

recall that: $\epsilon(x', x'') = i_A \epsilon'(x') + \lambda_P(x'')$. It's trivial on P'_0, one shall verify on P''_0:

$\eta F_0(0, x'') = \eta(\gamma_0(x''), F''_0(x'')) = i_B \circ \eta' \circ \gamma_0(x'') + \lambda_Q \circ F''_0(x'') = f \circ \epsilon(0, x'') = f \circ \lambda_P(x'')$

$\Rightarrow i_B \eta' \lambda_0 = f\lambda_P - \lambda_Q F''_0$

It's also a question of lift $\Rightarrow$ first check the image.

$\pi_B(f\lambda_P - \lambda_Q F_0'') = f''\pi_A\lambda_P - \pi_B\lambda_Q F_0'' = f''\epsilon'' - \eta'' F_0'' = 0$

$\Rightarrow im(f\lambda_P - \lambda_Q F_0'') \subseteq \ker \pi_B = im\, i_B \Rightarrow \exists \beta: P_0'' \to B' \text{ s.t. } i_B\beta = f\lambda_P - \lambda_Q F_0''$

$\Rightarrow \exists \gamma_0$:

$$\begin{array}{ccc} & & P_0'' \\ & \swarrow & \downarrow \beta \\ Q_0' & \xrightarrow{\eta'} & B' \longrightarrow 0 \end{array}$$

③ Assume $F_0, \cdots, F_{n-1}$ is constructed. Now we construct F_n $(n \geq 1)$

$\Rightarrow$ Verify it commutes w/ the differential $d_n^P = \begin{pmatrix} d_n^{P'} & \xi_n^P \\ & d_n^{P''} \end{pmatrix}$. $d_n^Q = \begin{pmatrix} d_n^{Q'} & \xi_n^Q \\ & d_n^{Q''} \end{pmatrix}$

$$\left[\begin{pmatrix} d' & \xi \\ & d'' \end{pmatrix}, \begin{pmatrix} F' & \gamma \\ & F'' \end{pmatrix}\right] = \begin{pmatrix} d'F' - F'd & (d'\gamma + \xi F'' - F'\xi - \gamma d'') \\ & d''F'' - F''d'' \end{pmatrix} = 0$$

$\Rightarrow d_n^{Q'}\gamma_n = \gamma_{n-1}d_n^{P''} - \xi_n F_n'' + F_{n-1}'\xi_n =: g_n : P_n'' \to Q_{n-1}'$

$$\begin{array}{ccc} P_n'' & & \\ \exists \gamma_n \downarrow & \searrow^{g_n} & \\ Q_n' & \xrightarrow{d_n^{Q'}} & Q_{n-1}' \end{array}$$

we shall check $im\, g_n \subseteq im\, d_n^{Q'} = \ker d_{n-1}^{Q'}$ (we shall not use $g_n = d_n^{Q'}\gamma_n$ to show $d_{n-1}^{Q'}g_n = 0$, since γ_n hasn't been constructed)

$\Rightarrow d_{n-1}^{Q'}(\gamma_{n-1}d_n^{P''} - \xi_n F_n'' + F_{n-1}'\xi_n)$

$= (\gamma_{n-2}d_{n-1}^{P''} - \xi_{n-1}F_{n-1}'' + F_{n-2}'\xi_{n-1})d_n^{P''} - d_{n-1}^{Q'}\xi_n F_n'' + d_{n-1}^{Q'}F_{n-1}'\xi_n$.

$= -\xi_{n-1}\underbrace{F_{n-1}''d_n^{P''}}_{=-d_n^{Q''}F_n''} + F_{n-2}'\xi_{n-1}d_n^{P''} - d_{n-1}^{Q'}\xi_n F_n'' + \underbrace{d_{n-1}^{Q'}F_{n-1}'\xi_n}_{=F_{n-2}'d_{n-1}^{P'}\xi_n}$. Since F'', F' is the chain map.

$= F_{n-2}'\underbrace{(\xi_{n-1}d_n^{P''} + d_{n-1}^{P'}\xi_n)}_{=0 \text{ from } d^P \circ d^P = 0} - (\xi_{n-1}d_n^{Q''} + d_{n-1}^{Q'}\xi_n)F_n'' = 0$ □

⑤ It's universal

Theorem 2.4.7 *Assume that $\mathcal{A}$ has enough projectives. Then for any right exact functor $F: \mathcal{A} \to \mathcal{B}$, the derived functors L_nF form a universal δ-functor.*

We shall prove: Given any δ-functor $\{T_n\}$ w/ $\varphi_0: T_0 \to F \Rightarrow \exists!\, \varphi_n: T_n \to L_nF$. $n > 0$ that commutes

Ⓐ We first define φ_n for every element.

Use induction, assume $\varphi_i: T_i \to L_iF$ is well-defined for $0 \leq i < n$, $n \geq 1$ and satisfy commuting relation

Given $A \in \mathcal{A}$, choose a projective P: $0 \to K \xrightarrow{i} P \to A \to 0$ is exact.

$$\begin{array}{ccccc} \Rightarrow T_n(A) & \xrightarrow{\delta_n^T} & T_{n-1}(K) & \xrightarrow{T_{n-1}(i)} & T_{n-1}(P) \\ \vdots \exists!\varphi_n^A & & \downarrow \varphi_{n-1}^K & & \downarrow \varphi_{n-1}^P \\ 0 \to L_nF(A) & \xrightarrow{\delta_n^L} & L_{n-1}F(K) & \xrightarrow{L_{n-1}(i)} & L_{n-1}F(P) \end{array}$$

i.e. $L_nF(A) = \ker L_{n-1}(i)$

Since: $L_{n-1}(i)\varphi_{n-1}^K\delta_n^T = \varphi_{n-1}^P \underbrace{T_{n-1}(i)\delta_n^T}_{=0} = 0$

From the universal property of kernel

$\Rightarrow \exists!\, \varphi_n^A: T_n(A) \to L_nF(A)$ s.t. $\delta_n^L\varphi_n^A = \varphi_{n-1}^K\delta_n^T$

③ Now we verify that φ_n is a natural transformation, which will reproduce that

$$\begin{array}{ccccc} T_n(K) & \to & T_n(P) & \to & T_n(A) \\ \downarrow \varphi_n & & \downarrow \varphi_n & & \downarrow \varphi_n \\ L_nF(K) & \to & L_nF(P) & \to & L_nF(A) \end{array}$$

commutes.

Given $f: A' \to A$, we shall prove that $L_nF(f)\varphi_n(A') = \varphi_n(A)T_n(f)$

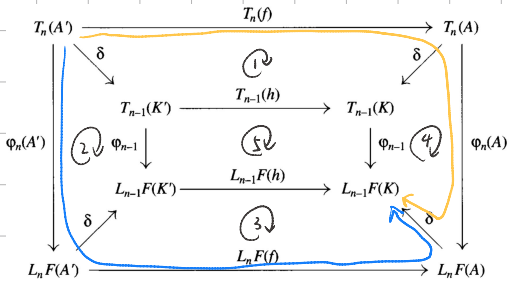
Use the SES $0 \to K' \to P' \to A' \to 0$ w/ P' projective.

$$\Rightarrow \begin{array}{ccccccc} 0 & \to & K' & \to & P' & \to & A' \to 0 \\ & & \downarrow h & & \downarrow g & & \downarrow f \\ 0 & \to & K & \to & P & \to & A \to 0 \end{array}$$

where g comes from the lift of f.

h comes from the universal property of kernel.

Assume that for $\varphi_0, \cdots, \varphi_{n-1}$, it's verified $(n \geq 1)$



①: T_* is a δ-functor

②④: just proven

③: L_*F is a δ-functor.

⑤: induction assumption.

Since $\delta_n^L L_nF(f)\varphi_n(A')$ ③ ② ⑤ ① ④ $\delta_n^L \varphi_n(A) T_n(f)$

and $0 \to L_nF(A) \xrightarrow{\delta_n^L} L_{n-1}F(K) \xrightarrow{L_{n-1}F(i)} L_{n-1}F(P) \Rightarrow \delta_n^L$ monic

$\Big\}\ L_nF(f)\varphi_n(A') = \varphi_n(A)T_n(f)$

② then we go back to general SES $0 \to A' \to A \to A'' \to 0$

use a SES $0 \to K'' \to P'' \to A'' \to 0$ w/ P'' projective

$$\Rightarrow \begin{array}{ccccccc} 0 & \to & K'' & \to & P'' & \to & A'' \to 0 \\ & & \downarrow g & & \downarrow f & & \| \\ 0 & \to & A' & \to & A & \to & A'' \to 0 \end{array} \Rightarrow \begin{array}{ccccc} T_n(A'') & \xrightarrow{\delta_n^L} & T_{n-1}(K'') & \xrightarrow{T(g)} & T_{n-1}(A') \\ \downarrow \varphi_n & & \downarrow \varphi_{n-1} & & \downarrow \varphi_{n-1} \\ L_nF(A'') & \xrightarrow{\delta_n^L} & L_{n-1}F(K'') & \xrightarrow{LF(g)} & L_{n-1}F(A') \end{array}$$

δ for $0 \to K'' \to P'' \to A'' \to 0$

$L_nF(A'') \xrightarrow{\delta_n^L} L_{n-1}F(K'')$

δ for $0 \to A' \to A \to A'' \to 0$ $\quad \searrow LF(g)$

$L_{n-1}F(A')$ ∎

Rm. the naturality of δ-functor guarantees that

Exercise 2.4.5 ([Tohoku]) An additive functor $F: \mathcal{A} \to \mathcal{B}$ is called *effaceable* if for each object A of $\mathcal{A}$ there is a monomorphism $u: A \to I$ such that $F(u) = 0$. We call F *coeffaceable* if for every A there is a surjection $u: P \to A$ such that $F(u) = 0$. Modify the above proof to show that if T_* is a homological δ-functor such that each T_n is coeffaceable (except T_0), then T_* is universal. Dually, show that if T^* is a cohomological δ-functor such that each T^n is effaceable (except T^0), then T^* is universal.

⇐ General property that:

$0 \to K \to P \to A \to 0$,

$L_nF(P) = 0, \ n \geq 1$

for the lifting property of P,

one shall use pullback

(omitted)

Right Derived Functor

let $F: \mathcal{A} \to \mathcal{B}$ be left exact. $\mathcal{A}$ has enough injectives $\Rightarrow$ construct the right derived functors:

Given $A \in \mathcal{A}$, choose an injective resolution $A \to I^\bullet$, the def:

$$R^i F(A) = H^i(F(I^\bullet))$$

Rm. ① $0 \to F(A) \to F(I^0) \to F(I^1)$ is exact $\Rightarrow R^0F(A) = \ker d^0 = F(A)$

② $F^{op}: \mathcal{A}^{op} \to \mathcal{B}^{op}$ is right exact $\Rightarrow L_iF^{op}$, one have: $R^iF(A) = (L_iF^{op})^{op}(A)$

Therefore all the results about right exact functors apply to left exact functors. In particular, the objects $R^iF(A)$ are independent of the choice of injective resolutions, R^*F is a universal cohomological δ-functor, and $R^iF(I) = 0$ for $i \neq 0$ whenever I is injective. Calling an object Q *F-acyclic* if $R^iF(Q) = 0\ (i \neq 0)$, as in 2.4.3, we see that the right derived functors of F can also be computed from F-acyclic resolutions.

③ For contravariant one:

e.g. let $F: \mathcal{A} \to \mathcal{B}$ contravariantly left exact, i.e. covariantly left exact $F: \mathcal{A}^{op} \to \mathcal{B}$

i.e. $0 \to A' \to A \to A'' \to 0$ in $\mathcal{A} \Leftrightarrow 0 \to A'' \to A \to A' \to 0$ in $\mathcal{A}^{op}$

$$0 \to F(A'') \to F(A) \to F(A')$$

Therefore, if $\mathcal{A}$ has enough projectives (i.e. $\mathcal{A}^{op}$ has enough injectives)

$\Rightarrow$ define right derived functor $R^iF(A) = H^i(P^\bullet)$ $P^\bullet \to A$

e.g. Since $\mathrm{Hom}_R(A, -)$ is left exact $\Rightarrow$ def: Ext functor $\mathrm{Ext}^i_R(A, B) = R^i\mathrm{Hom}_R(A, -)(B)$

Exercise 2.5.1 Show that the following are equivalent.

1. B is an *injective* R-module.
2. $\mathrm{Hom}_R(-, B)$ is an exact functor.
3. $\mathrm{Ext}^i_R(A, B)$ vanishes for all $i \neq 0$ and all A (B is $\mathrm{Hom}_R(-, B)$-acyclic for all A).
4. $\mathrm{Ext}^1_R(A, B)$ vanishes for all A.

The behavior of Ext with respect to the variable A characterizes projectives.

Exercise 2.5.2 Show that the following are equivalent.

1. A is a *projective* R-module.
2. $\mathrm{Hom}_R(A, -)$ is an exact functor.
3. $\mathrm{Ext}^i_R(A, B)$ vanishes for all $i \neq 0$ and all B (A is $\mathrm{Hom}_R(-, B)$-acyclic for all B).
4. $\mathrm{Ext}^1_R(A, B)$ vanishes for all B.

Since $- \otimes_R B$ is right exact $\Rightarrow$ def: Tor functor: $\mathrm{Tor}^R_n(A, B) = (L_n(- \otimes_R B))(A)$

Adjoint Functors & Left/Right Exactness

Thm. Let $L:\mathcal{A}\rightleftarrows\mathcal{B}:R$, i.e. there is a natural isomorphism $\tau:\mathrm{Hom}_{\mathcal{B}}(L(A),B)\xRightarrow{\cong}\mathrm{Hom}_{\mathcal{A}}(A,R(B))$. then $\Rightarrow$ L is right exact, R is left exact.

Proof Suppose that $0\to B'\to B\to B''\to 0$ is exact in $\mathcal{B}$. By naturality of τ there is a commutative diagram for every A in $\mathcal{A}$.

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathrm{Hom}_{\mathcal{B}}(L(A),B') & \longrightarrow & \mathrm{Hom}_{\mathcal{B}}(L(A),B) & \longrightarrow & \mathrm{Hom}_{\mathcal{B}}(L(A),B'') \\ & & \downarrow\cong & & \downarrow\cong & & \downarrow\cong \\ 0 & \longrightarrow & \mathrm{Hom}_{\mathcal{A}}(A,R(B')) & \longrightarrow & \mathrm{Hom}_{\mathcal{A}}(A,R(B)) & \longrightarrow & \mathrm{Hom}_{\mathcal{A}}(A,R(B'')) \end{array}$$

The top row is exact because $\mathrm{Hom}(LA,-)$ is left exact, so the bottom row is exact for all A. By the Yoneda Lemma 1.6.11,

$$0\to R(B')\to R(B)\to R(B'') \text{ is exact}$$

Dually: $L:\mathcal{A}\to\mathcal{B}$ left adjoint $\Leftrightarrow L^{op}:\mathcal{A}^{op}\to\mathcal{B}^{op}$ right adjoint $(\mathrm{Hom}_{\mathcal{B}^{op}}(B,L^{op}(A)))$

$\Rightarrow L^{op}$ is left exact $\Rightarrow L$ is right exact

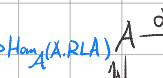
the natural isomorphism acturally corresponds to the existence & uniqueness of factorization.

Rm. Given $L:\mathcal{A}\rightleftarrows\mathcal{B}:R$, i.e. we have an isomorphism $\Phi_{A,B}:\mathrm{Hom}_{\mathcal{B}}(LA,B)\simeq\mathrm{Hom}_{\mathcal{A}}(A,RB)$

we gonna prove the factorization property:

$\forall A\in\mathcal{A}, B\in\mathcal{B}, \forall\alpha:A\to RB$, $\Phi_{A,LA}:\mathrm{Hom}_{\mathcal{B}}(LA,LA)\xrightarrow{\cong}\mathrm{Hom}_{\mathcal{A}}(A,RLA)$

$\Rightarrow\exists!\, g:LA\to B$ s.t. $\alpha=R(g)\circ\eta_A$, where $\eta_A:=\Phi_{A,LA}(id_{LA}):A\to RLA$ (is called the unit)



pf: Since $\alpha\in\mathrm{Hom}_{\mathcal{A}}(A,RB)\Rightarrow\exists!\, g\in\mathrm{Hom}_{\mathcal{B}}(LA,B)$ s.t. $\Phi_{A,B}(g)=\alpha$

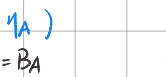
From the naturality: $\forall k:B\to B'$ in $\mathcal{B}$, $h:LA\to B$, $\Phi_{A,B'}(k\circ h)=R(k)\circ\Phi_{A,B}(h)$

then: $\alpha=\Phi_{A,B}(g)=\Phi_{A,B}(g\circ id_{LA})=R(g)\circ\Phi_{A,LA}(id_{LA})=R(g)\circ\eta_A$

Coro. Fix $R:\mathcal{B}\to\mathcal{A}$, it exists left adjoint iff

$\forall A\in\mathcal{A}, \exists B_A\in\mathcal{B}, \eta_A:A\to R(B_A)$ s.t. $\forall B\in\mathcal{B}, \alpha:A\to R(B)$, $\exists!\, h:B_A\to B$ s.t. $\alpha=R(h)\circ\eta_A$

(On the construction of the left adjoint once giving the unit η_A)



pf of ($\Leftarrow$): we want to show that B_A forms a functor: $L(A)=B_A$

$\Rightarrow$ need to verify $L(u)$ is well-defined for $u:A\to A'$

combine: $A\xrightarrow{u}A'\xrightarrow{\eta_{A'}}R(B_{A'})$, use the universal property $\Rightarrow\exists!\, L(u):B_A\to B_{A'}$

Finally: def: $\mathrm{Hom}_{\mathcal{B}}(LA,B)\to\mathrm{Hom}_{\mathcal{A}}(A,RB)$, $h\mapsto R(h)\circ\eta_A$

factorization $\Rightarrow$ surjective; uniqueness $\Rightarrow$ injective.

eg.

Application of adjoint: let R be a ring, B a left R-module $\Rightarrow - \otimes_R B \dashv \mathrm{Hom}_{Ab}(B, -)$

$\Rightarrow$ more generally: let B be a R-S bimodule $\Rightarrow (- \otimes_R B : \mathrm{mod}\text{-}R \to \mathrm{mod}\text{-}S) \dashv \mathrm{Hom}_{\mathrm{mod}\text{-}S}(B, -)$